# 18 Root Resorption

SHANON PATEL, CONOR DURACK, DOMENICO RICUCCI, and ABDULAZIZ A.BAKHSH

CHAPTER OUTLINE

**General Histologic Features**
**External Inflammatory Resorption**
Introduction
Etiology and Pathogenesis of External Inflammatory Resorption
Histologic Appearance
Clinical Features
Radiographic Features
Management
Follow-Up and Prognosis of External Inflammatory Resorption
**External Cervical Resorption**
Introduction
Etiology and Pathogenesis
Histologic Appearance
Clinical Features
Radiographic Features
Management
**Internal Root Resorption**
Introduction
Etiology and Pathogenesis of Internal Root Resorption
Histologic Appearance
Clinical Features
Radiographic Features and Diagnosis
Management
Chemomechanical Debridement of the Root Canal
Obturation
**Summary**

Dental resorption is defined as the loss of dental hard tissues as a result of clastic activities.[84] Root resorption in the primary dentition is indeed a physiologic process except if it occurs prematurely.[21,22] It is not yet completely understood how this process occurs, although it is believed that it is regulated by cytokines and transcription factors, which are also involved in bone remodeling.[54,125] On the other hand, resorption of permanent dentition is a pathologic process that, if left untreated, may result in premature loss of the affected teeth.

Depending on its location in relation to root surface, root resorption may be classified as external and internal.[4,107] Internal root resorption (IRR) was reported as early as 1830.[18] The etiology and pathogenesis of IRR have not been completely demonstrated.[64,88] Diagnosis of IRR is often a concern to the clinician because it is often confused with external cervical resorption (ECR). An inaccurate diagnosis may result in inappropriate treatment in certain cases.[51,52,83,87,88]

## General Histologic Features

Bone resorption is a process regulated by motile, multinucleated giant cells called osteoclasts. They are formed by the fusion of mononuclear precursor cells of the monocyte-macrophage lineage derived from the spleen or bone marrow; osteoblasts and osteocytes, on the other hand, are derived from skeletal precursor cells.[72,103] Osteoclasts are recruited to the site of injury or irritation by the release of many proinflammatory cytokines. To perform their function, osteoclasts must attach themselves to the bone surface. On contact with mineralized extracellular matrices, the actin cytoskeleton of an actively resorbing osteoclast is reorganized to produce an organelle-free zone of sealing cytoplasm (clear zone) associated with the osteoclast's cell membrane; this enables the osteoclast to achieve intimate contact with the hard tissue surface.[91] The clear zone surrounds a series of fingerlike projections (podosomes) of cell membrane, known as the *ruffled border,* beneath which bone resorption occurs. The resorptive area within the clear zone therefore is isolated from the extracellular environment, creating an acidic microenvironment for the resorption of hard tissues.

Odontoclast cells are morphologically similar to osteoclasts but they are responsible of resorbing dental hard tissue (Fig. 18.1).[44] Odontoclasts differ from osteoclasts in that they are smaller, have fewer nuclei, and have smaller sealing zones, possibly as result of differences in their respective resorption substrata.[66] Osteoclasts and odontoclasts resorb their target tissues in a similar manner.[91] The two cells have similar enzymatic properties,[79] show similar cytologic features, and create resorption depressions, termed *Howship lacunae*, on the surface of the mineralized tissues (see Fig. 18.1).[91] Odontoclasts are polarized in relation to dental tissue and have a ruffled border, located inside a clear zone, that is in intimate contact with their dental substratum.[60,91]

At high-power magnification, the cytoplasm of the odontoclasts appears vacuolated and shows a more intense staining reaction than the cytoplasm of the adjacent cells. Where odontoclasts are in contact with the tooth surface, the characteristic brush border can be seen, but the hematoxyphilic zone lining the lacunae is absent.[44] When odontoclasts are observed with an electron microscope, the most striking features are the large number of mitochondria and vacuoles in the cytoplasm and the scarcity of endoplasmic reticulum. An

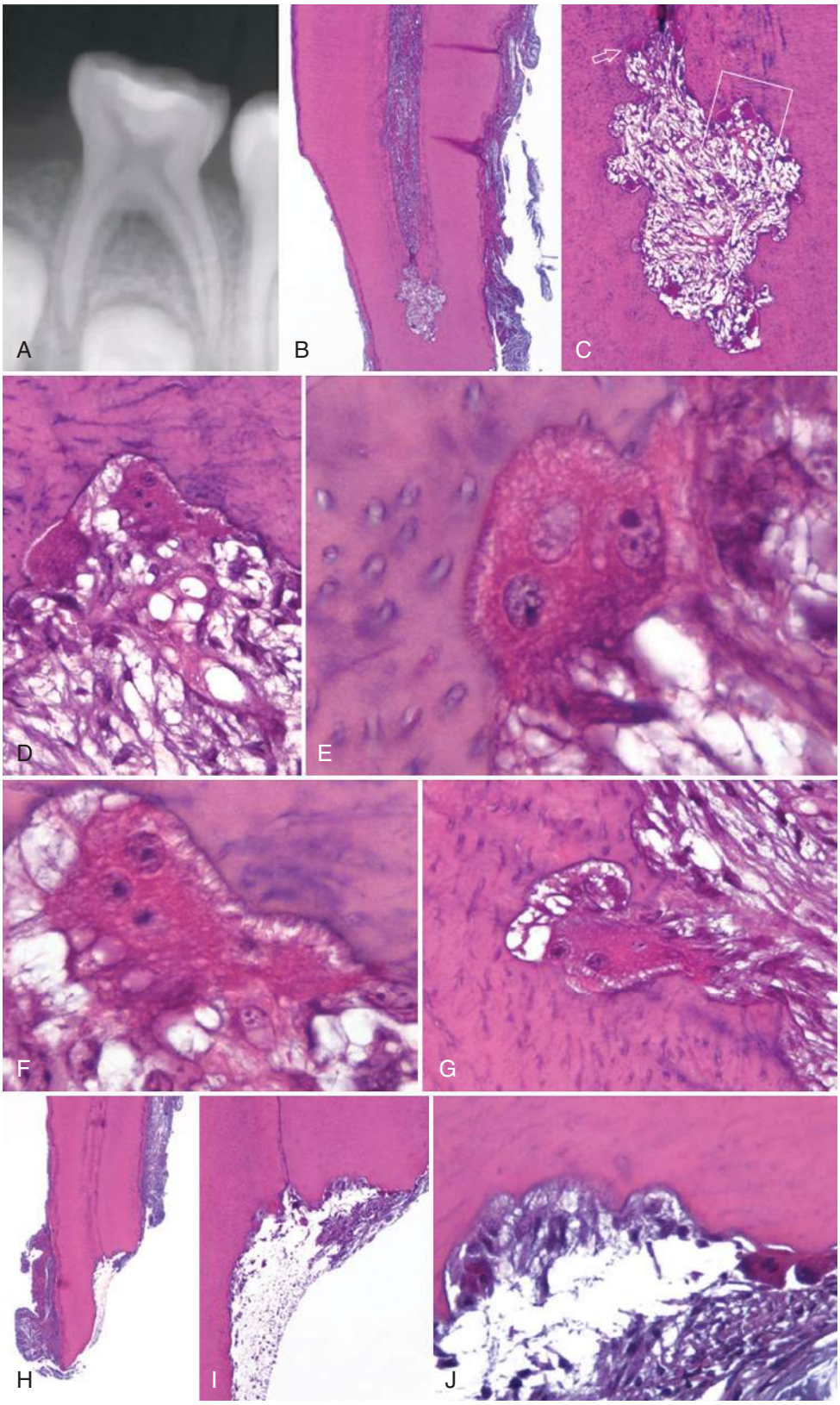



**Fig. 18.1** Light microscopy images of a deciduous tooth showing both physiologic and pathologic (inflammatory) resorption. **A,** Tooth 7.5 in a 9-year-old boy. Extensive exposure of a mesial pulp horn occurred during caries excavation, and pulp capping was performed. The patient was brought to the dental office 4 months after the treatment because of severe spontaneous pain. The radiograph shows radiolucencies on both apices and interradicularly. The parents did not accept any treatment and requested extraction. **B,** Longitudinal section taken on a mesiodistal plane, passing approximately at the center of the mesial root. (Hematoxylin-eosin [H&E] stain; original magnification ×25.) **C,** Detail of the resorption area apically in **B** (original magnification ×100). **D,** Magnification of the area demarcated by the rectangle in **C.** Two odontoclasts are present in an area of dentin being resorbed, surrounded by fibroblasts and inflammatory cells (original magnification ×400). **E,** High-power view of the area indicated by the *arrow* in **C.** An odontoclast is in close contact with the dentin in a Howship lacuna. Its cytoplasm appears vacuolated and shows a more intense staining reaction than the cytoplasm of the adjacent cells (original magnification ×1000). **F,** High-power view of the upper odontoclast in **D.** The characteristic ruffled border can be distinguished (original magnification ×1000). **G,** Howship lacuna with an odontoclast in another area (original magnification ×400). **H–J,** Progressive magnification of the mesial root apex. The resorption process may be physiologic at this level (original magnification ×16, ×100, and ×400).

abundance of cytoplasmic ribosomes can be seen. The nucleoli are fairly large and centrally located in the nucleus. Close to the nucleus, the Golgi apparatus appears as a narrow zone of fine canals. Where the odontoclasts are in contact with the tooth surface, the ruffled border (corresponding to the brush border) may be observed with the light microscope. The ruffled border consists of cytoplasmic folds that create a system of canals extending 2 to 3 μm into the cytoplasm. In undecalcified sections, mineral crystals could be observed in these canals.[44]

Although both mononuclear dendritic cells and multinucleated osteoclasts share a common hematopoietic lineage, they have previously been accounted for as immunologic defense cells. Recent studies have indicated that immature dendritic cells also function as osteoclast precursors and have the potential to transdifferentiate into osteoclasts.[104] Because dendritic cells are present in the dental pulp, it is possible they also may function as precursors of odontoclasts.

From a molecular signaling perspective, the osteoprotegerin (OPG)/receptor activator of nuclear factor κB ligand (RANKL)/RANK transcription factor system[23] that controls clastic functions during bone remodeling has also been identified in root resorption.[111] The system is responsible for the differentiation of clastic cells from their precursors by means of complex cell-cell interactions with osteoblastic stromal cells. Similar to periodontal ligament (PDL) cells that are responsible for external root resorption,[114] the human dental pulp has recently been shown to express OPG and RANKL messenger ribonucleic acids (mRNAs).[112] OPG, a member of the tumor necrosis factor superfamily, has the ability to inhibit clastic functions by acting as a decoy receptor that binds to RANKL and reduces its affinity for RANK receptors on the surface of clastic precursors. This results in inhibition of the regulation of clastic cell differentiation. Thus, it is possible that the OPG/RANKL/RANK system may be actively involved in the differentiation of odontoclasts during root resorption.

We know that osteoclasts do not adhere to nonmineralized collagen matrices.[108] It has been suggested that the presence of a noncollagenous, organic component in dentin (odontoblast layer and predentin) prevents (internal) resorption of the root canal wall, and precementum prevents (external) resorption of the external root surface.[117,119] Similar to osteoclasts, odontoclasts may bind to extracellular proteins containing the arginine-glycine–aspartic acid (RGD) sequence of amino acids by means of integrins.[100] The latter are specific surface adhesion glycoprotein membrane receptors containing different α and β subunits. In particular, $\alpha_v\beta_3$ integrin plays a key role in the adhesion of clastic cells.[76] Extracellular matrix proteins containing the RGD peptide sequence, particularly osteopontin, present on the surface of mineralized tissues and serve as binding sites of clastic cells.[58] The osteopontin molecule contains different domains, with one domain binding to apatites in the denuded dentin and another domain binding to integrin receptors in the plasma membranes of clastic cells. Thus, osteopontin serves as a linker molecule that optimizes the attachment of a clastic cell to mineralized tissues, mediating the rearrangement of its actin cytoskeleton.[30] It has been speculated that the lack of RGD peptides in predentin reduces the binding of odontoclasts, thereby conferring resistance of the canal walls to IRR.

Once the clastic cells have established contact with the cementum, any subsequent resorption is self-limiting unless the bound cells are subject to continued stimulation.[4] Therefore, in addition to the precipitating events discussed previously, progressive root resorption requires a source of stimulation for the resorbing cells. Stimulating factors vary and are related to the site and type of resorption, in addition to the cause of the predisposing damage to the predentin or precementum. Examples include persistent pressure and forces associated with continued orthodontic treatment; persistent impacted teeth; untreated cysts, granulomas, and tumors; endodontic inflammation and/or infection; and periodontal inflammation and/or infection.[45]

## External Inflammatory Resorption

### INTRODUCTION

External inflammatory resorption (EIR) affects the outer surface of the root and is usually a consequence to dental luxation[2] and avulsion[13] injuries. It is a progressive condition with a potentially precipitous onset, and it is capable of advancing rapidly, such that an entire root surface may be resorbed within a few months if the tooth is left untreated.[5,13] It also affects teeth diagnosed with chronic periapical periodontitis (Figs. 18.2–18.5).[62]

After luxation injuries, EIR ranges from almost 5%[2] to 18%.[32] It affects 30% of replanted avulsed teeth.[10,11] EIR is the most common form of external resorption root resorption after luxation and avulsion injuries.[32]

Treatment of EIR should be carried out as soon as it is diagnosed, and it is mainly based on effective removal of the infected necrotic pulp.[37] The earlier the diagnosis, management, and treatment of EIR, the better the prognosis of the affected tooth.[35]

The diagnosis of EIR in clinical situations is based solely on the radiographic demonstration of the process.[3] In some cases, the initial radiographic signs of EIR can be visualized as early as 2 weeks after replantation of avulsed teeth.[13] However, the limitations of conventional radiographic imaging in dentistry have been well reported. The diagnostic yield of radiographs is reduced by adjacent anatomic noise,[19,50,61,94] geometric distortion,[50] and compression of three-dimensional (3D) structures onto a two-dimensional (2D) shadowgraph.[31,82,116] These limitations may result in late diagnosis of EIR after dental trauma.

In a prospective study including up to 637 permanent luxated teeth in 400 patients for 10 years, Andreasen and Vestergaard Pedersen[2] reported that no cases of EIR after concussion injuries and only one case after subluxation injuries, which represented only 0.5% of the total number of subluxated teeth. Six percent of extrusively luxated teeth and 3% of laterally luxated teeth developed EIR. However, EIR was a healing complication in 38% of intrusive luxation injuries.[2] Crona-Larsson et al.[32] also reported a higher prevalence of EIR after more severe luxation injuries, with 60% of extrusively luxated teeth and 22% of intruded teeth affected by the complication. They reported EIR to be a much more frequent occurrence after lateral luxation (16.7%) and subluxation injuries (3.8%) relative to the findings of Andreasen and Vestergaard Pedersen.

### ETIOLOGY AND PATHOGENESIS OF EXTERNAL INFLAMMATORY RESORPTION

Normally, permanent teeth are resistant to resorption.[118,119,120] It is generally accepted that odontoclasts do not bind to the nonmineralized layers covering the external root surface and the root canal wall (i.e., the precementum and predentin,

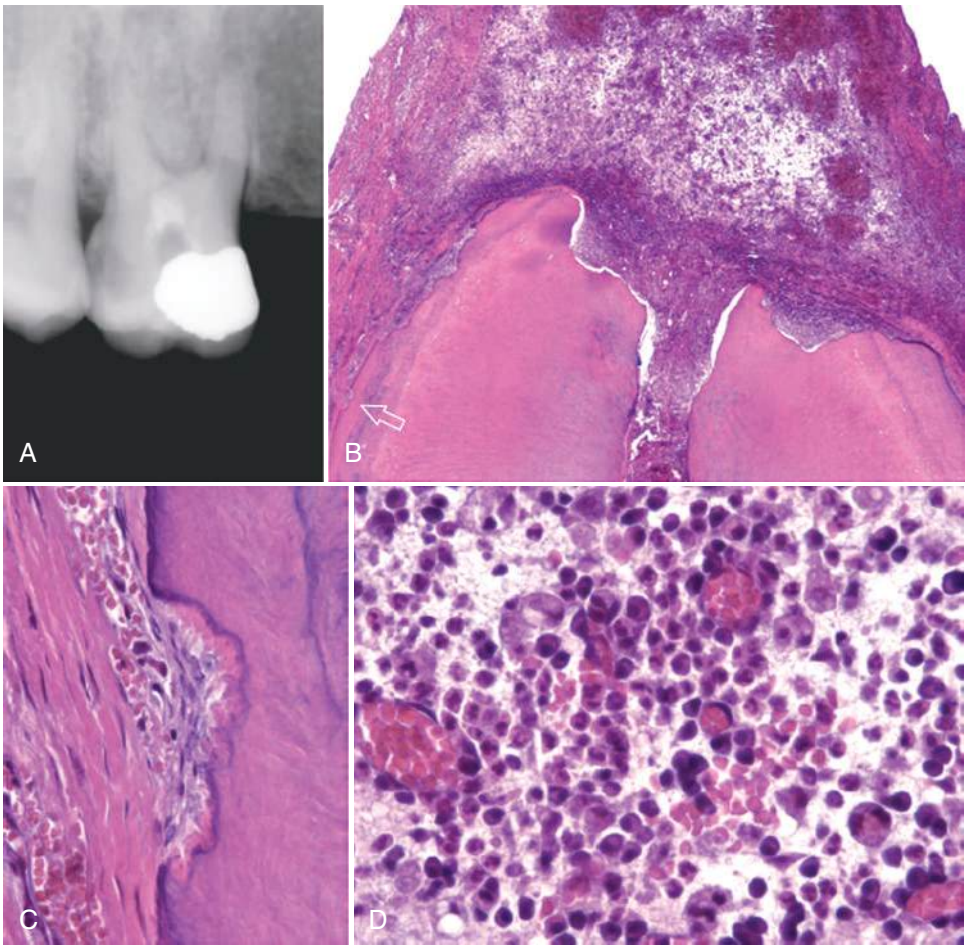



**Fig. 18.2** External apical inflammatory resorption. **A,** Maxillary first molar in a 46-year-old female patient that was causing severe pain. The radiographs revealed material in the pulp chamber, a large distal caries, and apical radiolucencies. The tooth was extracted. **B,** Palatal root apex extracted with the pathologic tissue attached. Note the ingrowth of granulation tissue into the foramen and the massive resorption of the apical profile. Also note the resorption of lacunae on the left radicular profile. (H&E stain; original magnification ×25.) **C,** High magnification of the area of the external radicular profile indicated by the *arrow* in **B.** A resorption lacuna can be seen in the cementum. (H&E stain; original magnification ×400.) **D,** View from the center of the apical periodontitis lesion showing severe concentration of chronic inflammatory cells (mostly plasma cells) (original magnification ×400).

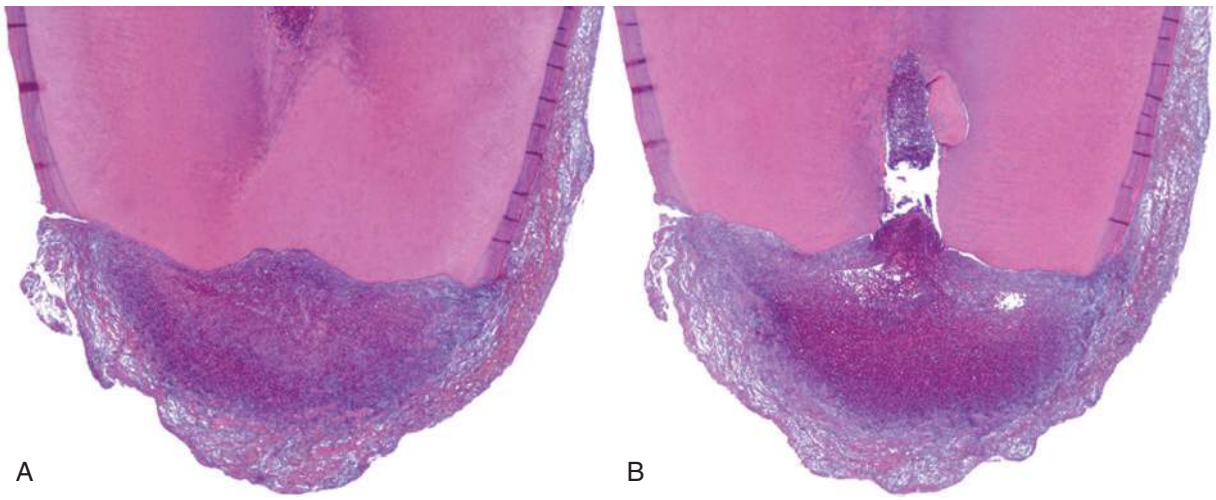



**Fig. 18.3** External apical inflammatory resorption. **A,** Mandibular second premolar extracted with the periapical lesion attached. This section, which did not pass through the canal, shows extensive apical resorption. (H&E stain; original magnification ×25.) **B,** Section taken approximately 120 sections away encompasses the apical foramen. In addition to resorption, the opposite phenomenon can be observed; that is, a large calcification partly embedded in the right apical dentin wall (original magnification ×25).

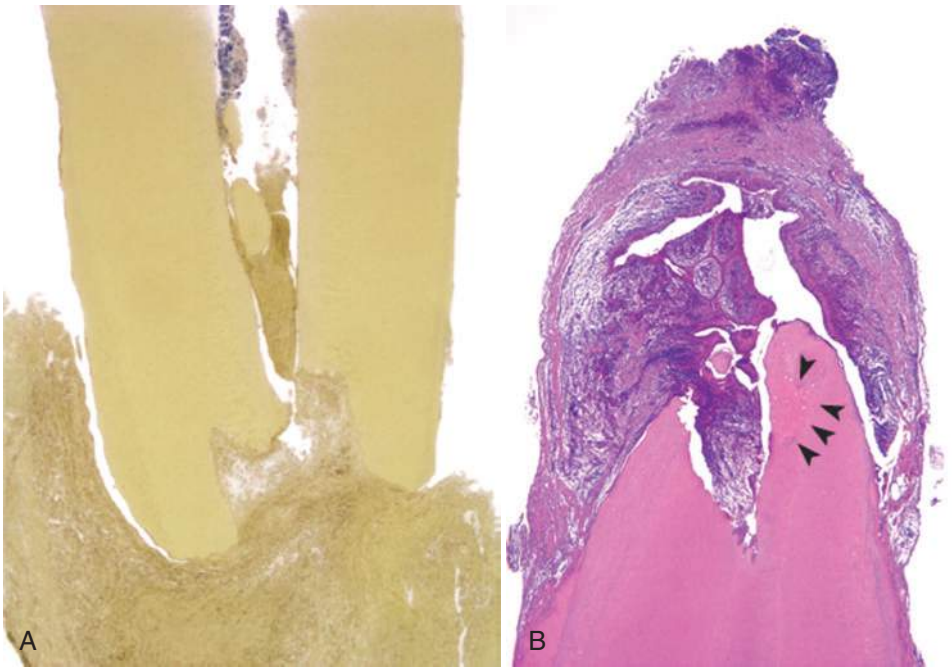



**Fig. 18.4** External apical inflammatory resorption. **A,** Mesial root of a mandibular first molar extracted with a periapical lesion attached. This section passes through the canal and the foramen. Extensive resorption of the foramen has occurred, and a calcification can be seen more coronally. Note that the tissue in the apical canal and in the resorptive defect appears structured and continuous with the apical periodontitis lesion. A thick biofilm can be discerned layering the root canal walls more coronally. (Taylor's modified Brown & Brenn stain; original magnification ×25.) **B,** Palatal root of a maxillary first molar extracted with the periapical lesion attached. A cyst cavity is present in the body of the lesion. Resorption of the foramen is present, and a portion of the right dentin wall has been replaced by bone *(arrowheads).* (H&E stain; original magnification ×25.)

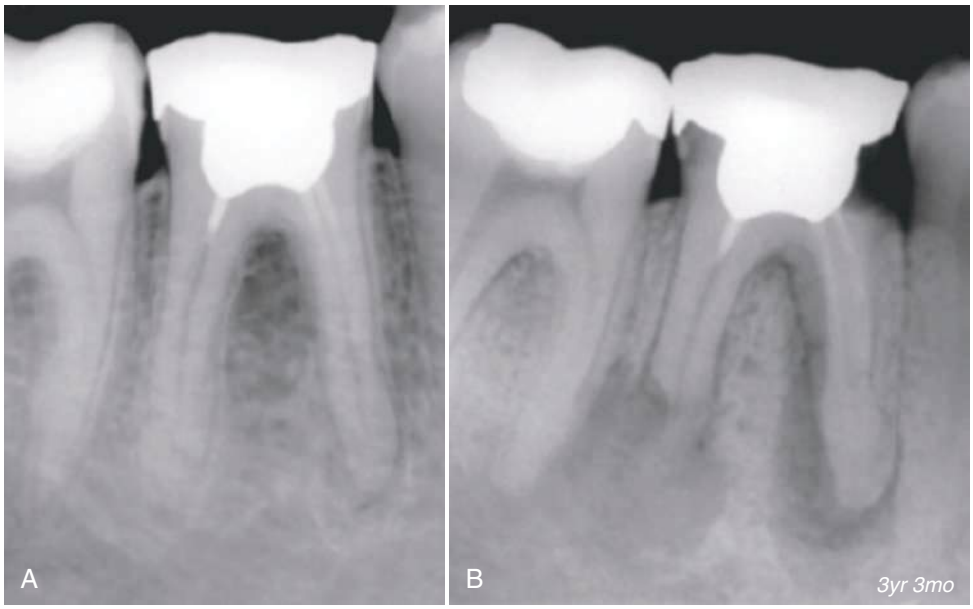



**Fig. 18.5** Apical resorption. **A,** 22-year-old male patient complained of pain on chewing. This was caused by a molar that had been treated endodontically 1 year earlier. A radiograph showed obturation material only at the root canal orifices and a periapical radiolucency on the mesial root. Endodontic retreatment was recommended, but the patient declined this treatment. **B,** The patient returned 3 years and 3 months later because of an abscess that was causing severe pain. A radiograph disclosed destructive mesial caries and large periapical radiolucencies on both roots, along with resorption of the whole apical third of the distal root.

respectively).[4,53,118,119,120] Similarly, it has been shown that, between episodes of physiologic resorption, the surface of bone is impervious to osteoclastic activity because it is covered by a layer of nonmineralized collagen fibrils, to which the osteoclast is incapable of binding.[29]

Traumatic dental injuries (e.g., intrusion, lateral luxation, and avulsion) and subsequent replantation often result in contusion injuries to the PDL.[4] Damage to the precementum, with a resultant breach in its integrity, is the precipitating factor in all types of external resorption.[4,6]

In the subsequent wound healing process, necrotic PDL tissue remnants are excavated and removed by macrophages and osteoclasts.[39] A critical factor is that the precementum may be stripped from the root surface during the injury and the damaged cementum and bone may also be phagocytosed, resulting in exposure of the underlying dentine to osteoclastic and odontoclastic activity.[39] External root resorption may ensue, but the precise type depends on the severity of the initial injury, the stage of root development, and the pulp status of the affected tooth.[4,6] Once the clastic cells have established contact with the cementum or dentine, any subsequent resorption is self-limiting unless the cells are subject to continued stimulation.[9] As already mentioned, odontoclasts do not adhere to the nonmineralized layers covering the external root surface.[4,53,118,119,120] Therefore, damage to the precementum with a resultant breach in the integrity of these layers is the precipitating factor in all types of ERR.[4,6] Damage to the precementum and underlying tooth structure allow circulating odontoclasts to bind to the underlying mineralized dentine and cementum.[118]

The pathogenesis of EIR can be explained as follows. Contusion injuries to the PDL, after a traumatic dental injury (TDI) involving the periodontal structures, initiate wound healing, during which osteoclasts and macrophages are attracted to the site of the injury to remove the damaged tissue. The initial injury causes a breach in the integrity of the protective precementum. This permits odontoclasts to bind to and resorb the underlying mineralized cementum and dentine in a manner similar to the development of surface resorption.[6] However, EIR differs from surface resorption in that it is a progressive event that relies on microbial stimulation from the infected necrotic pulp of the affected tooth for its progression.[9] Therefore, it is more commonly associated with tooth avulsions[10,11] and moderate to severe luxation injuries, which have the potential to compromise pulp vitality.[2] If the strength of the initial osteoclast attack is sufficient to expose patent dentinal tubules beneath the cementum, a communication is created between the pulp space and the external root surface and adjacent periodontal tissues.[9,14] Microbes and/or their toxins (e.g., lipopolysaccharide, muramyl dipeptide, and lipoteichoic acid),[15] located in the root canal and the dentinal tubules, diffuse through the tubules and directly stimulate the resorbing osteoclasts. The resorption process intensifies and accelerates.[7] The resorbed mineralized tissue is replaced by granulation tissue, which ultimately invades the pulp space if the process continues.[15]

Pressure on the root surface during orthodontic treatment[63] and from impacted teeth,[124] cysts,[106] and tumors[65] may also denude the protective precementum from the root surface and therefore initiate ERR.

## HISTOLOGIC APPEARANCE

The histologic appearance of EIR is characterized by saucer- or bowl-shaped areas of resorption in both the cementum and dentine, with concomitant inflammation in the adjacent periodontal membrane. Howship lacunae are a common feature of the resorption cavities, and histologic sections show that the lacunae sometimes are occupied by odontoclasts. The inflammatory reaction in the periodontal membrane appears intense and consists of a mixed-cell infiltrate that includes plasma cells, lymphocytes, and polymorphonuclear leucocytes in a granulation tissue matrix. Proliferation of capillaries in the areas of inflammation is also a feature.[14] EIR can be identified histologically 1 week after experimental replantation of teeth.[8]

## CLINICAL FEATURES

The tooth in question will not respond to sensibility testing and in advanced cases may be associated with signs of pulpitis or apical periodontitis such as discoloration, sinus tract, and/or tenderness to percussion and/or palpation.

## RADIOGRAPHIC FEATURES

As mentioned, diagnosis of EIR is based solely on radiographic demonstration of the process.[3,13] EIR is characterized radiographically by radiolucent, concave, and sometimes ragged bowl-shaped excavations along the root surface, with corresponding and associated radiolucencies in the adjacent alveolar bone. Complete loss of the lamina dura is seen in the area of the resorption.[14] The initial radiographic signs of EIR can often be seen as early as 3 to 4 weeks after a TDI involving the periodontal tissues,[14] and if it will develop, EIR is always seen within 1 year after the injury.[7]

EIR can have a rapid onset and aggressive progression, such that complete resorption of an entire root can occur within 3 months. The diagnostic potential of a number of radiographic imaging systems has been investigated, with varying degrees of success.

Conventional intraoral radiographic imaging (digital or film based) is currently the clinical reference standard for the detection of ERR after luxation and avulsion injuries.[41,42] However, it has been well documented that this form of imaging is an inadequate method of detecting simulated ERR, especially when the cavity sizes are small.[3] Clinical studies have also demonstrated that conventional radiography grossly underestimates the extent of inflammatory root resorption.[40]

Clinical studies directly comparing the ability of intraoral radiographs and cone beam computed tomography (CBCT) to detect and diagnose EIR are limited. One clinical study reported that CBCT is superior to conventional radiography in diagnosing and determining the extent of nonspecific inflammatory resorption on root surfaces.[40] D'Addazio et al.[36] compared the ability of CBCT and periapical radiography to detect simulated external resorption cavities of about 2 mm in diameter in a human ex vivo model. Although both imaging modalities were 100% sensitive in the detection of the lesions, only CBCT could accurately assess the position of the defects on the root surface and their relationship to the root canal, even though multiple angled periapical radiographs of the test teeth were available to the examiners.

## MANAGEMENT

In cases were the tooth is treatable, root canal treatment will be able to help in arresting the resorption and therefore prevent further progression of the resorption. This would

create an environment conducive to hard tissue repair of the damaged root surface.[33,34,37] Therefore, it is essential to initiate root canal treatment as soon as radiographic signs of EIR are identified.[35] An exception to this is replanted teeth with closed apices; in these cases, root canal treatment should be carried out 7 to 10 days after replantation, even if there are no radiographic signs of EIR.[42] The earlier the resorption is diagnosed and treated, the better the prognosis is for the affected tooth. Failure to diagnose and treat the condition may result in tooth loss.

Effective chemomechanical debridement of the root canal space is fundamental to the success of the root canal treatment and the inhibition and cessation of EIR.[34,37] In principle, the specific root canal protocol used is irrelevant, as long as the biologic objectives are met. Long-term dressing of the root canal with calcium hydroxide may be beneficial in the treatment of established EIR; however, this protocol should be used judiciously because of the associated risk of root fracture.[12]

In many cases the EIR is extensive, rendering the tooth unsalvageable and requiring extraction.

### FOLLOW-UP AND PROGNOSIS OF EXTERNAL INFLAMMATORY RESORPTION

Healing of EIR is characterized radiographically by cessation of the resorption process, resolution of the radiolucency in the adjacent bone, and reestablishment of the PDL space.[13] As mentioned previously, in untreated cases, EIR can progress so rapidly that an entire root can be resorbed within 3 months.[13,34] The prognosis is especially poor for untreated immature teeth.[13]

## External Cervical Resorption

### INTRODUCTION

ECR manifest mainly in the cervical region and it may invade root dentin in any direction and to a varying degree. ECR generally develops immediately apical to the epithelial attachment of the tooth. In healthy teeth with a normal periodontal attachment, this is in the tooth's cervical region, a feature that gave rise to the name. However, in teeth that have developed gingival recession and lost periodontal support and/or have developed a long junctional epithelium, the resorptive defect may arise at a more apical location.

ECR has also been referred to as invasive cervical resorption,[56] supraosseous extracanal invasive resorption,[43] peripheral inflammatory root resorption,[48] and subepithelial external root resorption.[109] The authors of this chapter prefer the term ECR because it describes the nature and location of the lesions.

### ETIOLOGY AND PATHOGENESIS

The exact etiology and pathogenesis of ECR have not been fully elucidated. It is widely accepted that the resorptive process is the same for ECR as it is for any other type or resorption: a breach in the protective nonmineralized layers must exist to allow the clastic cells to bind to the underlying dentine, and the same cells must be stimulated to perpetuate the process. However, in ECR only some of the factors that predispose the root surface to clastic activity have been identified.

The anatomic profile of the cementoenamel junction (CEJ) is variable, and the junction between the enamel and the cementum in this region is not contiguous in all teeth. This may lead to exposed areas of unprotected dentin, which are vulnerable to osteoclastic activity, in the cervical region of some teeth.[78]

Mavridou et al.[69] investigated the potential predisposing factors using a two-step approach. The first step was an interview with the patient to obtain a complete information on their dental and medical history, the second step was a thorough clinical investigation of the patient with a combination of periapical radiograph and 3D CBCT scans.

Of the 347 teeth examined, 59% were multifactorial. From these multifactorial cases, 38% had 2 predisposing factors, and 17% and 3% had 3 and 4 factors, respectively.

Orthodontics was the most common appearing factor (45.7%) followed by trauma (28.5%), parafunctional habits (23.2%), poor oral hygiene (22.9%), malocclusion (17.5%), and extraction of neighboring tooth (14%).

Combination of orthodontics with previous dental trauma was the most common combination (17.6%); this was followed by orthodontics along with parafunctional habits (13%), orthodontics with extraction of neighboring tooth (12%), extraction of neighboring tooth along with occlusal overloading and poor oral health (5%), malocclusion with poor oral health (3%), and malocclusion along with parafunctional habits (2.7%).

Of the cases examined, ECR was located on 29% of maxillary central incisors, 14% in maxillary canines, 14% in mandibular molars, and 14% maxillary premolars. ECR was less frequently seen in mandibular canines (3%), mandibular centrals (1%), and mandibular lateral (1%).[69]

### HISTOLOGIC APPEARANCE

The histologic profile of ECR is similar to that of other forms of resorption, with certain unique features reflecting the invasive nature of the process.

Recently, histologic analysis has revealed that ECR lesion is a mechanism of a three-stage process. Initially, the ECR starts at the cementum apical to the gingival epithelium (i.e., the initiation stage). In the second stage of the resorption (i.e., the resorption phase), the resorptive lesion invades the tooth structure in a 3D manner towards the pulp space. However, the pulp is protected by a pericanalar resorption-resistant sheet which prevents the resorption lesion from penetrating the pulp space. This layer has a fluctuating thickness of 210 μm and consists of predentin, dentin, and occasionally bonelike tissue. At the last advanced stage (i.e., the repair stage), bonelike tissue is deposited into the resorption cavity in which repair takes place (Fig. 18.6).[70]

In a case study investigating the presence of hypoxia with ECR, Mavridou et al.[71] reported that histologic analysis showed that hypoxia does exist. Hypoxia inducible factor 1a-positive odontoblast was found within the apical area of the resorption (Figs. 18.7 and 18.8).

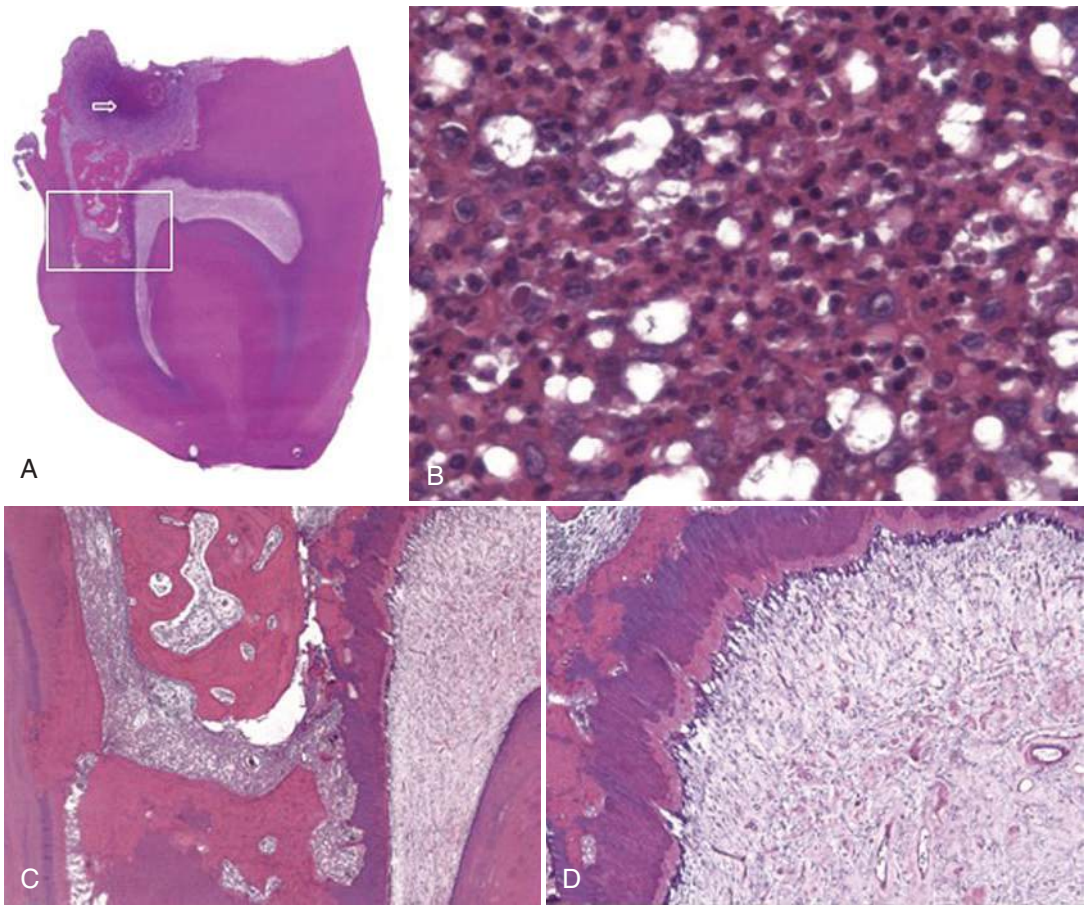



**Fig. 18.6** External cervical resorption with replacement. **A,** Asymptomatic mandibular third molar. ECR was diagnosed on a radiograph taken for neighboring teeth. The tooth was extracted. Overview of the tooth (H&E stain; original magnification ×2). **B,** High power view of the resorptive area occupied by granulation tissue indicated by the *arrow* in **A**. Severe concentration of acute and chronic inflammatory cells (original magnification ×400). **C,** Detail of the area demarcated by the *rectangle* in **A**. The large area of resorbed dentin has been partly repaired with bonelike tissue. Note the thin layer of dentin remaining. No inflammatory changes can be observed in the adjacent pulp (original magnification ×25). **D,** Detail from **C**. Normal appearance of dentin, predentin, and odontoblast layer. Absence of inflammation (original magnification ×50).

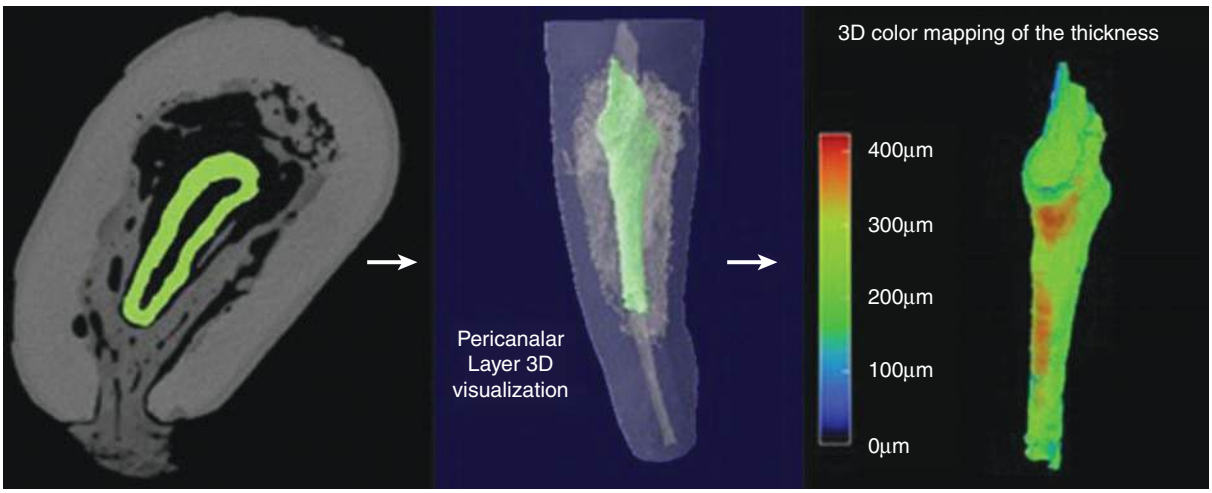



**Fig. 18.7** NanoCT imaging of tooth 13 and 3D modeling using CTan, CTvol, and CTvox software, showing the thickness distribution of the pericanalar resorption resistant sheet (PRRS). (From Patel S, Mavridou A. M, Lambrechts P, Saberi N: External cervical resorption-part 1: histopathology, distribution and presentation. *Int Endod J* 51:1205–1223, 2018.)

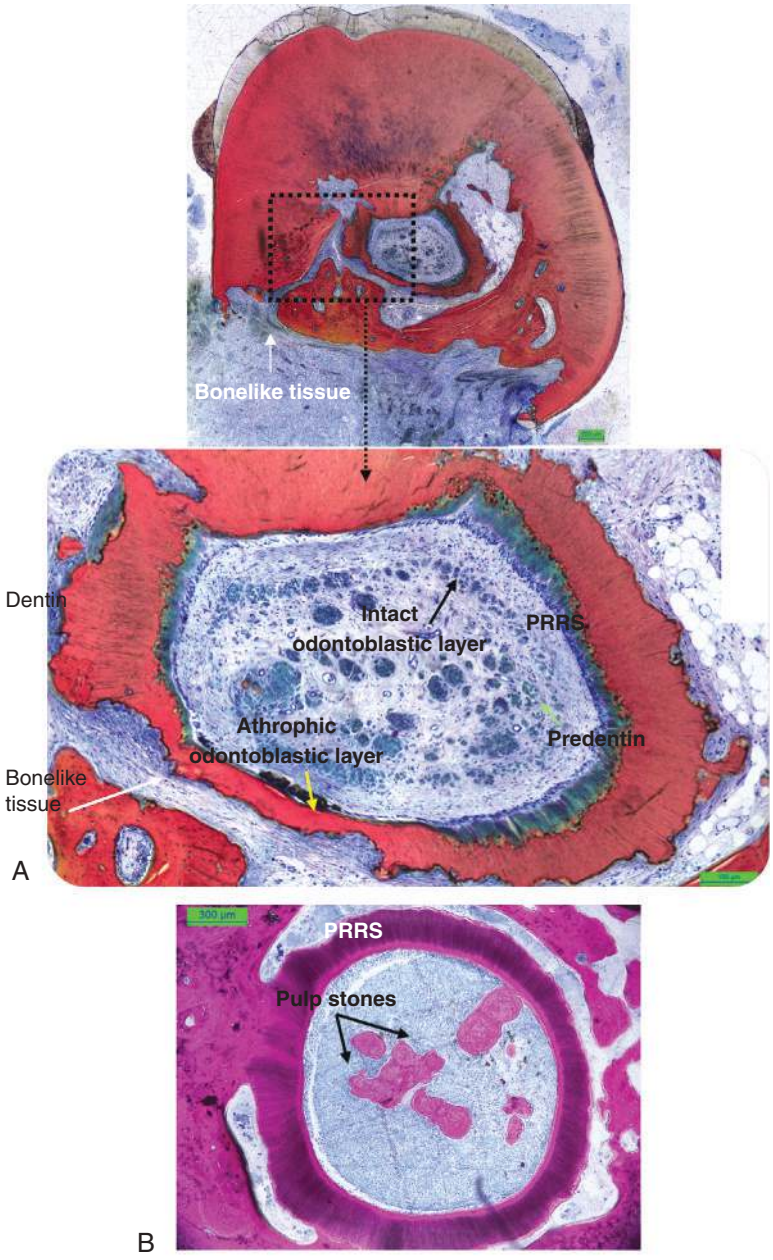



**Fig. 18.8 A,** Structure of pericanalar resorption resistant sheet *(PRRS)* and pulp tissue. PRRS is consisted of predentin, dentin and occasionally bonelike tissue. **B,** Calcification inside the pulp. Histologic staining was performed with a combination of Stevenel's blue and Von Gieson's picrofuchsin, visualizing mineralized tissue *(red)* and nonmineralized tissue *(blue-green)*. (From Patel S, Mavridou A. M, Lambrechts P, Saberi N: External cervical resorption-part 1: histopathology, distribution and presentation. *Int Endod J* 51:1205–1223, 2018.)

## CLINICAL FEATURES

The clinical features of ECR are variable (Figs. 18.9 and 18.10). The process is very often quiescent and asymptomatic, especially in the earlier stages, and absence of clinical signs and symptoms is very common; the diagnosis is commonly made as a result of a chance radiographic finding. A pink or red discoloration may develop at the cervical region of the tooth; when present, this often is the feature that alerts the patient or clinician to the possible existence of a problem. The discoloration is due to the fibrovascular granulation tissue occupying the resorptive defect, which has a reduced thickness of enamel and dentin at its peripheries because of the loss of hard tissue. The granulation tissue imparts a pink hue to the tooth, through the thinned enamel and dentin, in the region of the resorption.[84] The granulation tissue may perforate the enamel or dentin at the gingival margin, giving the appearance of mild gingival hyperplasia. The discoloration, sometimes referred to as a "pink spot," can be quite subtle and is often a chance finding by the patient, the dentist or, increasingly, the dental hygienist. However, it is a relatively rare feature of ECR. Furthermore, it must occur at a site where it is readily identifiable (e.g., labial surface of an anterior tooth) to be noticed. Loss of periodontal attachment may occur in the region of the resorption, and probing of the resorptive defect or the associated periodontal pocket causes the granulation tissue to bleed profusely.[87]

As the process progresses, perforation of the root canal wall and bacterial contamination of the pulp may occur. The affected tooth may develop pulpitis and the associated clinical symptoms. Pulp necrosis and chronic periapical periodontitis may eventually develop. Clinical signs and symptoms may be the first indication of a problem with the affected tooth;

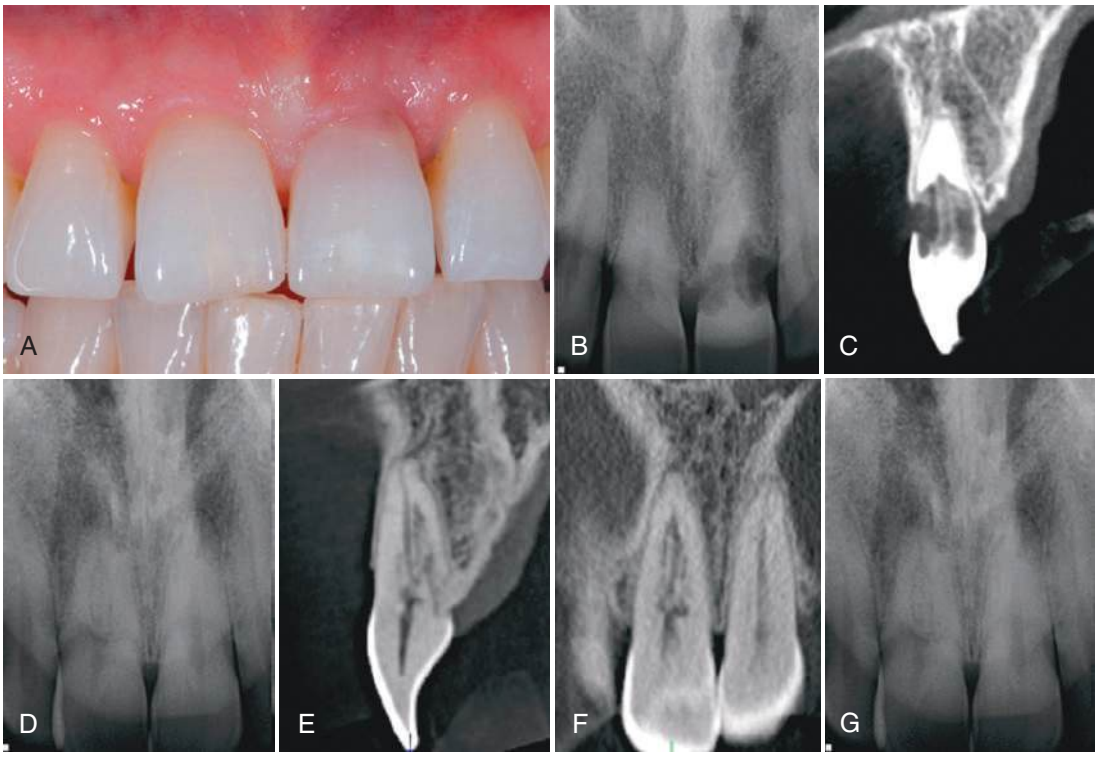



**Fig. 18.9** External cervical resorption (ECR). **A,** A 55-year-old female patient presented with an asymptomatic "pink spot." She had no history of any predisposing factors. **B,** A periapical radiograph revealed radiolucent defects on the proximal aspects of the upper left central incisor; note the ragged borders. **C,** A reconstructed coronal cone beam computed tomography (CBCT) slice reveals the true extent of the ECR lesion. Note that the root canal wall appears to be intact. Inhibitory factors in the root canal wall/odontoblastic prevented the ECR lesion from penetrating the root canal. **D,** A 41-year-old male patient presented as a new patient. Routine radiographic examination revealed a poorly defined periapical radiolucency in the root of the upper right central incisor. The appearance is suggestive of ECR; the patient had orthodontic treatment in his early teens and remembered "knocking the tooth" at least twice when he was very young. Reconstructed sagittal **(E)** and coronal **(F)** CBCT slices revealed the true nature of the ECR lesion and showed that the lesion was not amenable to treatment. **G,** The treatment options were discussed with the patient, and it was decided to review the tooth periodically. The 4-year radiograph shows no change in size of this asymptomatic lesion.

they may include tooth discoloration, spontaneous localized pain, tenderness on mastication, tenderness to percussion, tenderness to palpation over the apical region of the tooth, a draining sinus, and/or buccal sulcus swelling.

## RADIOGRAPHIC FEATURES

There is no usual radiographic appearance of ECR. The lesion may have well-defined or irregular margins in the cervical aspect of the tooth.

The radiographic appearance of ECR depends on the location, the extent of invasion, and the relative proportions of fibro-osseous and fibrovascular tissue occupying the resorptive cavity. All ECR defects present as a radiolucency of varying radiodensity, often in but not confined to the cervical region of the affected tooth or teeth (Figs. 18.11 and 18.12; also see Figs. 18.9 and 18.10).

If the lesion is predominantly fibrovascular and granulomatous, the lesion tends to be radiolucent. However, in long standing lesion, the radiolucency may appear with a cloudy appearance. In advanced cases with extensive repair of the tissue destruction, significant deposition of fibro-osseous tissue gives the defect a mottled radiographic appearance (Fig. 18.13).

The margins of the lesion may vary from poorly to well defined, depending on the depth of the defect and the proportion and distribution of osseous inclusions in the lesion. Although lesions with irregular margins are more common, some ECR defects may have smooth and/or well-defined margins.

Despite its often cervical location, ECR may commence apical to this region, reflecting the position of the epithelial attachment of the affected tooth. In teeth with a normal periodontal attachment, the invasive nature of the process may result in the lesion extending some distance apical and/or coronal to the cervical location where it started. Furthermore, the tissue destruction at the site of onset may sometimes be minimal and/or not evident on conventional radiographs due to its location on the root surface. In these instances, the lesion may appear to have originated at a location where significant tissue destruction, evident radiographically, has occurred. This may be some distance from the actual point of origin. This feature has come to light only since the advent of assessment of ECR with CBCT.

The radiographic features of ECR are very similar to those of IRR (discussed later), and differentiating between them, especially in the absence of clinical signs, may be challenging. It is useful to trace the outline of the root canal walls as they approach and pass through the resorption defect on the radiograph. In cases of ECR, the outline of the canal wall should be visible and intact and should maintain its course as it passes through the defect. This is due to the fact that the resorptive lesion lies on the external surface of

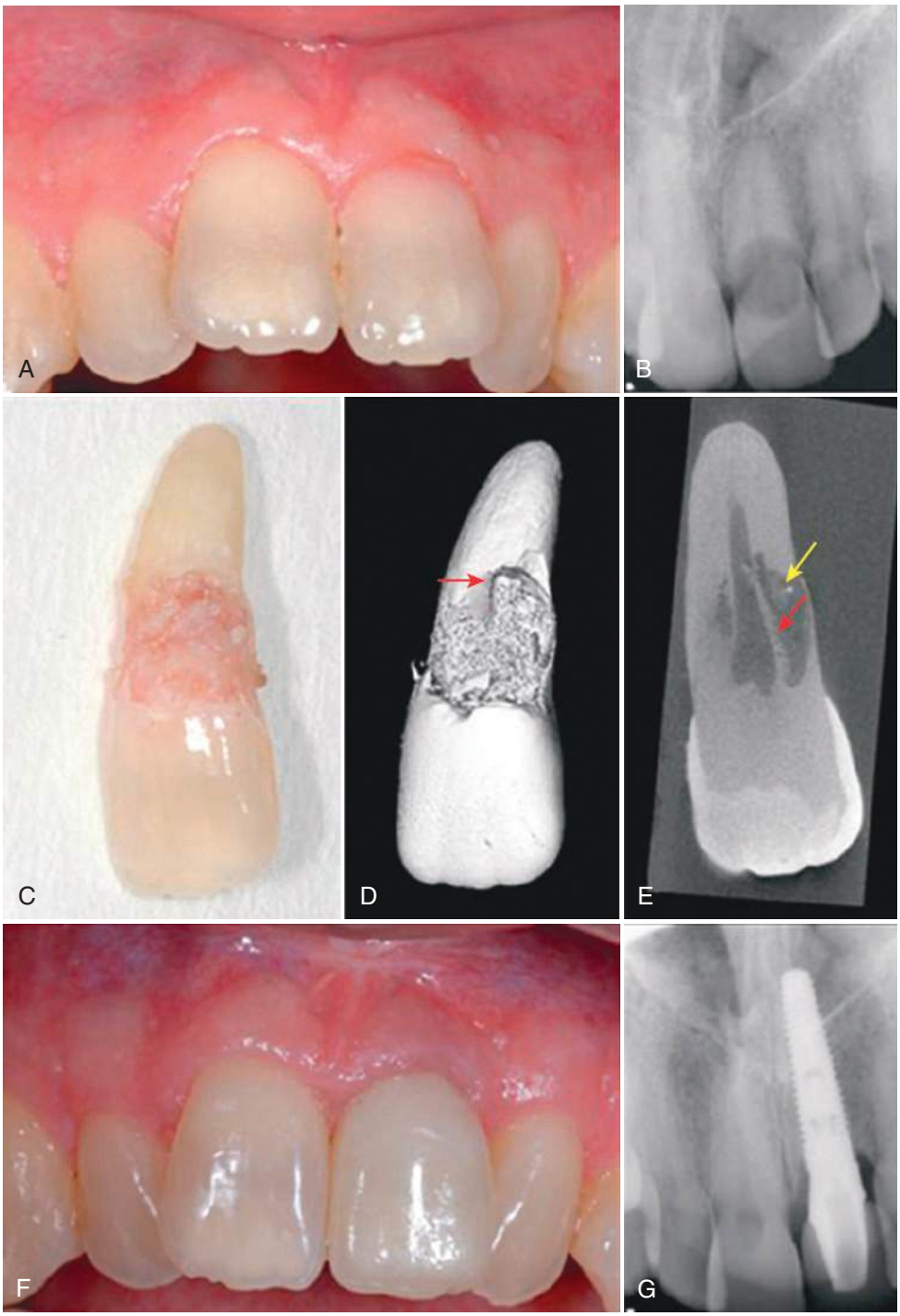



**Fig. 18.10 A,** This patient's upper left central incisor had a pink spot and was cavitated. **B,** A radiograph revealed an unusual presentation of ECR; the lesion is circular and has well-defined margins. Note that the outline of the root canal is visible and intact through the radiolucent lesion. **C,** The tooth was unrestorable and was extracted; note the large amount of granulation tissue. **D,** Three-dimensional reconstruction of the extracted tooth from microtomography data revealed bonelike tissue below the overlying granulation tissue. Note the intact root canal wall *(red arrow)*. **E,** Coronal reconstruction from a microtomography scan reveals how the predentin *(red arrow)* prevented the ECR defect from invading the root canal. In addition, bonelike tissue can be seen *(yellow arrow).* Posttreatment view **(F)** and radiograph **(G)** after replacement of the upper left central incisor with an implant-retained crown. (From Patel S, Kanagasingam S, Pitt Ford T: External cervical resorption: a review. *J Endod* 35:616–625, 2009.)

the root and is not in communication with the root canal; it is merely superimposed on the defect radiographically. In cases of IRR, it should be possible to trace the outline of the root canal through the resorptive defect because the defect is an extension of the root canal wall and is continuous with it.[46] Although this is a useful diagnostic feature, it does have some shortcomings. First, the outline of the root canal wall may be obscured by calcified tissue in the resorptive defect (ECR or IRR). Second, when ECR has resulted in extensive tissue destruction, perforation of the root canal wall

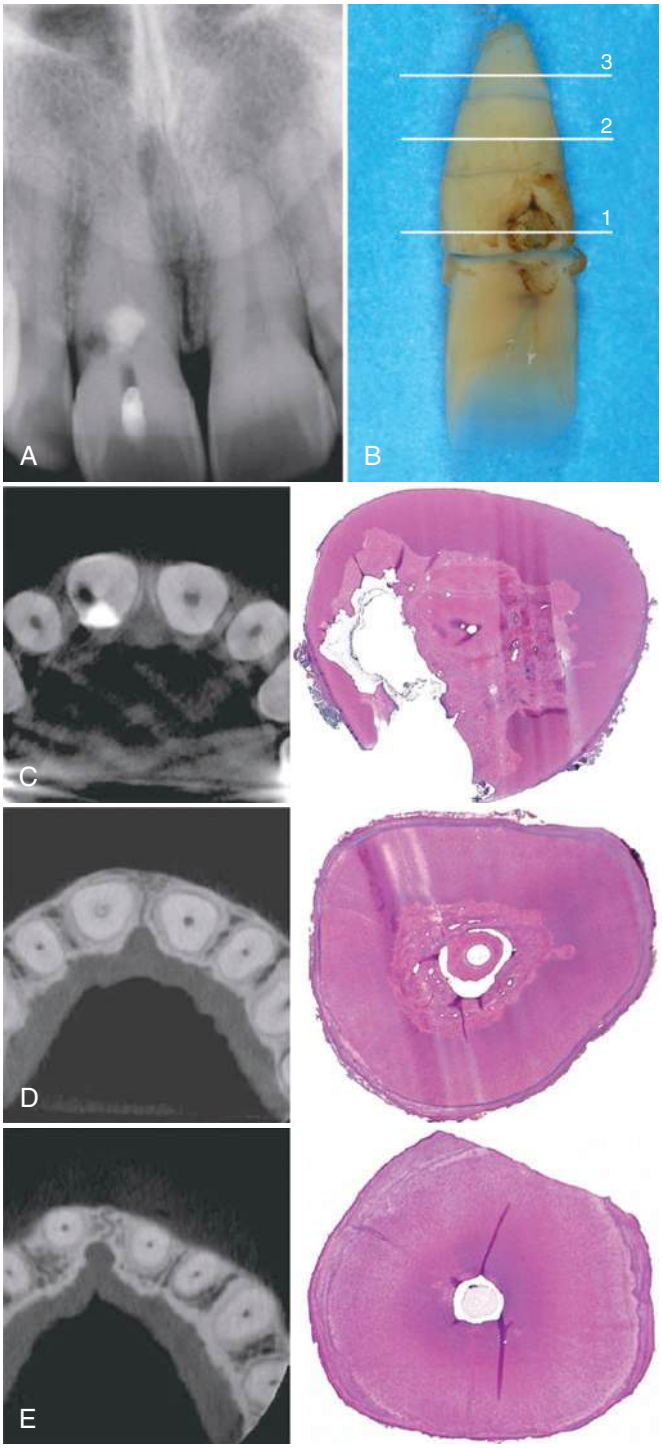



**Fig. 18.11** External cervical resorption (ECR) with replacement. **A,** Radiograph of a maxillary central incisor of a 34-year-old female patient who recalled being struck in the face with a cricket ball when she was 11 years old. The dentist mistakenly diagnosed the resorptive defect as caries and attempted to manage it accordingly. The tooth was asymptomatic; however, a 4-mm periodontal probing depth was identified on the palatal aspect of the tooth. A cone beam computed tomography (CBCT) scan confirmed an ECR defect. The tooth was deemed unrestorable and was extracted with the patient's consent. **B,** Palatal view of the tooth at the end of the demineralization process, while immersed in the clearing agent. The tooth was separated into four portions, which were embedded separately in paraffin blocks. **C,** CBCT axial section passing through the coronal third at the level of *line 1* in **B.** The corresponding histologic section shows that most of the dentin has been replaced by a bonelike tissue. The root canal is no longer present. (H&E stain; original magnification ×8.) **D,** CBCT axial section passing through the middle third at the level of *line 2* in **B.** The root canal can be appreciated at this level, although reduced in size, but it appears to be encircled by bonelike tissue (original magnification ×8). **E,** CBCT axial section taken from the apical third at the level of *line 3* in **B.** At this level, the canal is the same size of that of the contralateral tooth. The histologic section confirmed the absence of bonelike tissue (original magnification ×8).

may have allowed communication between the canal wall and the external defect.

Parallax radiographs should always be used to obtain further information about the nature of the resorptive process. In addition to a paralleled periapical radiograph, another radiograph should be taken with a shift (parallax) in the horizontal angulation of the x-ray tube in relation to the image receptor. In cases of ECR, the position of the resorptive defect moves relative to the root canal. If the lesion is located palatally/lingually, the defect moves in the same direction as the x-ray tube shift. If the lesion is located buccally, it moves in the opposite direction. This is sometimes referred to as the "same lingual, opposite buccal" (SLOB) rule. In contrast, internal resorptive defects maintain their position relative to the root canal because the defects are an extension of the root canal system.

CBCT has allowed 3D assessment of the nature, position, and extent of the resorptive defect, eliminating diagnostic confusion and providing essential information about the restorability and subsequent management of the tooth. CBCT is particularly useful if the clinician is not sure whether the ECR cavity has perforated the root canal wall

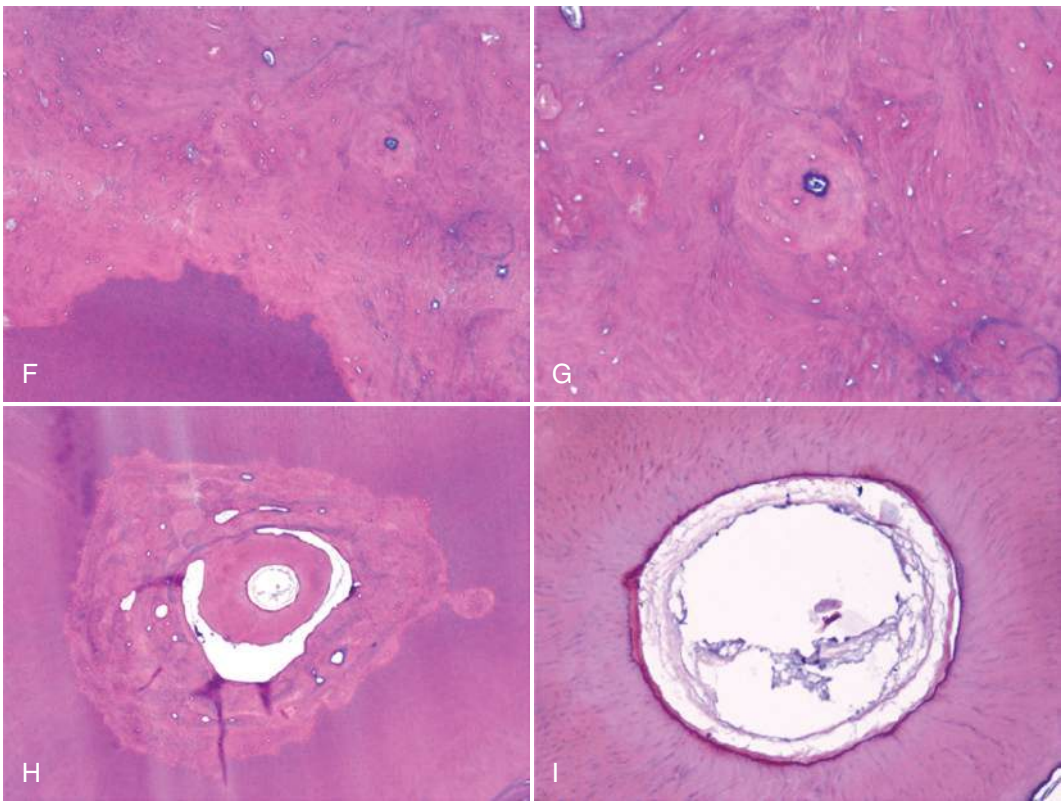



**Fig. 18.11, cont'd** **F,** Detail from **C** showing the transition from dentin to bonelike tissue (original magnification ×100). **G,** Higher magnification of **F.** The metaplastic tissue does not show the lamellar structure typical of bone (original magnification ×400). **H,** Detail of the canal in **D.** The metaplastic bonelike tissue has concentrically replaced a consistent portion of dentin, leaving in place a reduced layer of the original dentin (original magnification ×25). **I,** High-power view confirms that the tissue surrounding the canal is dentin (original magnification ×400). *Considerations:* The resorption process, followed by replacement with a bonelike tissue, started in the cervical area and extended in an apical direction, tunneling dentin circumferentially up to the transition between the middle and apical thirds of the tooth.

(and thus for determining the need for root canal treatment). A CBCT scan eliminates the need for exploratory treatment. In an ex vivo study, Vaz de Souza et al.[113] compared the diagnostic efficacy of two CBCT scanners with periapical radiograph for the detection and classification of simulated ECR lesions. Examiners were asked to classify and identify the root surface(s) that ECR extended to with periapical radiograph and CBCT. Identification of the lesion according to the tooth surface was successful in 89.1% Morita CBCT, 87.8% Kodak CBCT, and 49.4% periapical radiograph cases. According to Heithersay classification, ECR was classified correctly in 71.4% Kodak CBCT, 70% Morita CBCT, and 32% periapical radiograph.

A clinical study by Patel et al.[83] comparing the ability of conventional radiographs and CBCT to diagnose and to differentiate accurately between IRR and ECR showed that CBCT was significantly more accurate (100%) than periapical radiographs at diagnosing the presence and nature of the root resorption. These researchers also concluded that the correct treatment plan should incorporate the additional information provided by CBCT.[83]

Recently, Patel et al.[86] suggested a 3D classification that accounts for the lesion height, circumferential spread, and proximity to the root canal. The aim of the classification is to ensure accurate diagnosis and communication of ECR between clinicians (Table 18.1).

## MANAGEMENT

The fundamental treatment objectives in ECR are to excavate the resorptive defect, halt the resorptive process, restore the hard tissue defect with an esthetic filling material, and prevent and monitor the tooth for recurrence. Effective management of ECR depends on accurate assessment of the true nature and accessibility of the lesion.

Preoperative CBCT scans can help in assessing the extent of the defect and aid in good treatment planning. Also, treatment options could be confidently discussed with the patient and "exploratory" treatment may be avoided.[97,98]

Small lesions have the most favorable outcome where typically the pulp is not involved. These lesions are considered as Heithersay class 1 and class 2.[55] Surgical access to the site of resorption is gained by raising a mucoperiosteal flap, the dimensions of which should allow visualization of the full extent of the defect. Once access has been achieved, the resorptive cavity is excavated. Fibrovascular granulomatous tissue is readily removed with a hand excavator. However, defects containing significant amounts of fibro-osseous tissue (especially when the latter is contiguous with the adjacent dentin) require discriminate removal of the tissue with ultrasonic instruments. It may be extremely difficult to differentiate between sound dentin and fibro-osseous deposits; therefore, use of the surgical operating microscope is essential. Frequent intraoperative radiographs may be

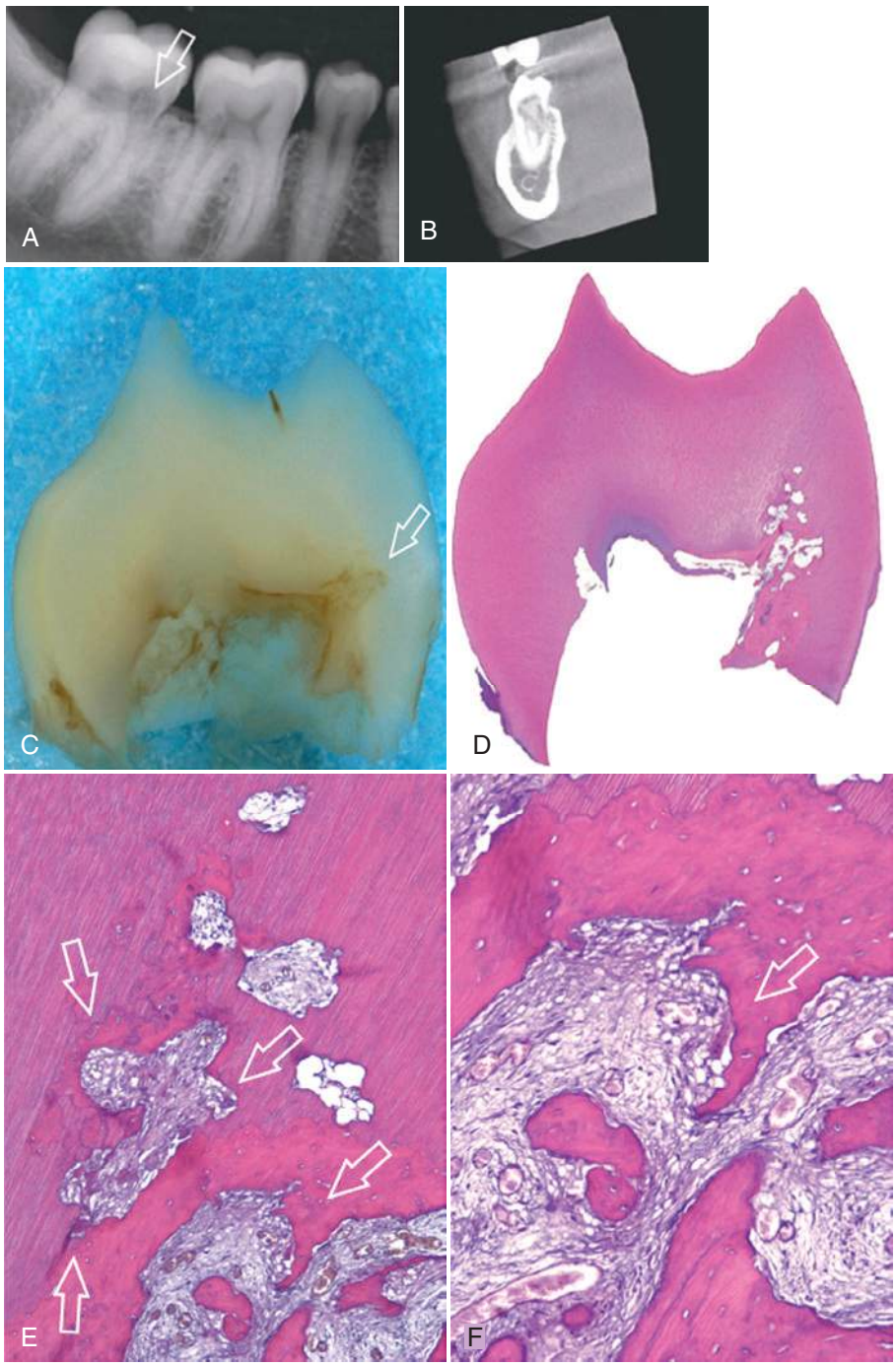



**Fig. 18.12** External cervical resorption with replacement. **A,** Mandibular first and second molars in a 27-year-old female patient. The patient had symptoms of irreversible pulpitis associated with the lower right second molar; the lower right first molar was asymptomatic. **B,** CBCT scan taken through the second molar at the level of the area indicated by the *arrow* in **A.** Note massive resorption involving the crown and root. **C,** Mesial portion of the crown of the second molar after clearing. **D,** Section taken on a buccolingual plane. Overview shows resorption and replacement on the lingual side, corresponding to the area indicated by the *arrow* in **C.** (H&E stain; original magnification ×6.) **E,** Detail of the area indicated by the *arrow* in **C.** Several areas of resorption, with replacement by metaplastic tissue, can be seen (original magnification ×100). **F,** Magnification of the area indicated by the right *lower arrow* in **E.** The metaplastic tissue closely resembles bone (original magnification ×400).

necessary to ensure accurate removal of unwanted, hard resorptive tissue and to prevent unnecessary removal of sound dental tissue.

Once the resorptive tissue has been removed, the cavity may be treated with a 90% aqueous solution of trichloroacetic acid; this causes coagulation necrosis of the resorptive tissue without damaging the periodontal tissue.[55] The acid also penetrates and treats small channels of resorption that are not accessible to mechanical instrumentation.[87] Once the defect has been excavated and treated with trichloroacetic acid, any undermined dentin or enamel at the peripheries of the cavity is removed with a bur in a high-speed turbine, and the cavity is restored with an esthetically acceptable restorative material, such as composite resin or

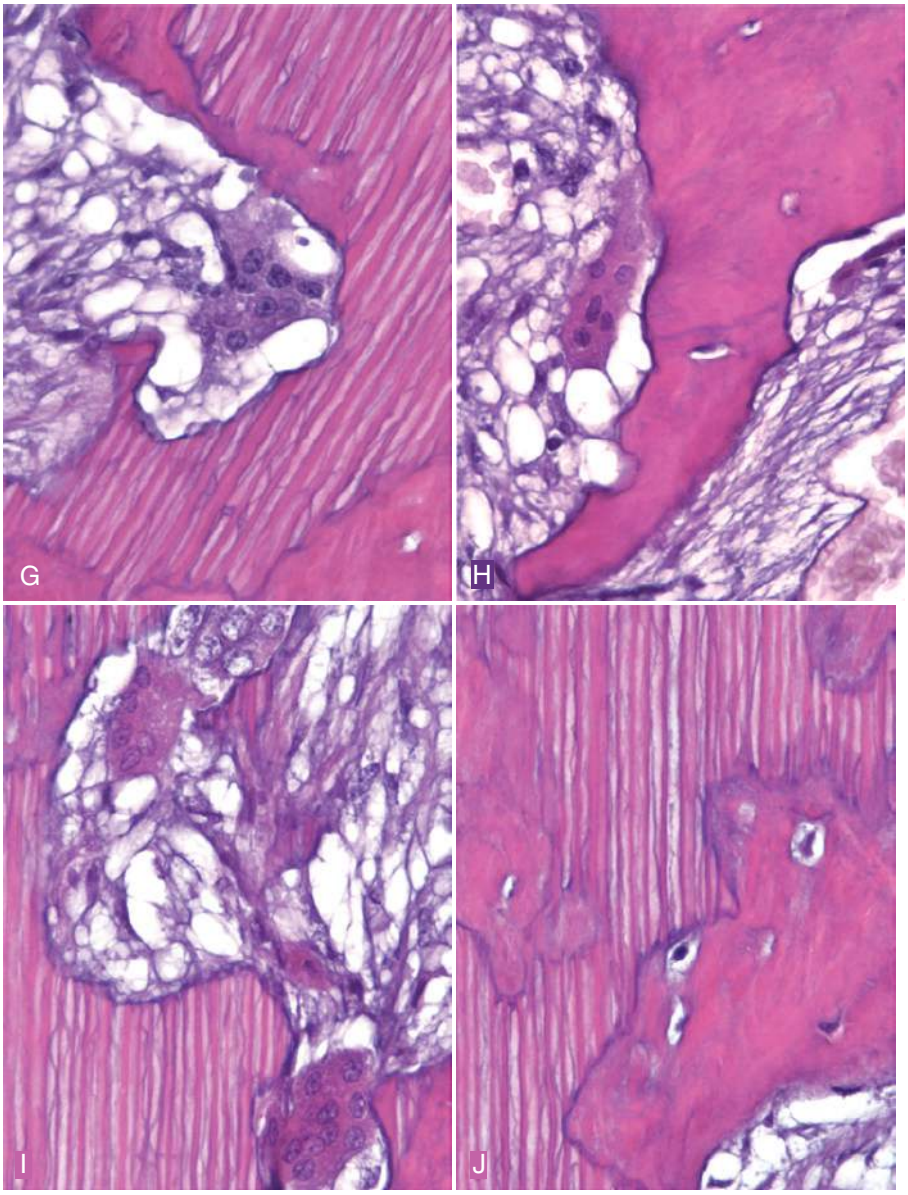



**Fig. 18.12, cont'd** **G,** High-power view of the area indicated by the right *upper arrow* in **E.** Note the lacuna in dentin occupied by a multinucleated clastic cell (original magnification ×1000). **H,** High-power view of the metaplastic bone tissue in the area indicated by the *arrow* in **F.** A bone trabecula is being resorbed by a typical osteoclast (original magnification ×1000). **I,** High-power view of the area indicated by the left *lower arrow* in **E.** A Howship lacuna with odontoclasts can be seen (original magnification ×1000). **J,** High-power view of the area indicated by the left *upper arrow* in **E.** Note the islands of bone tissue, with typical osteocytes, surrounded by dentin (original magnification ×1000). *Considerations:* In this case the metaplastic tissue was similar to normal lamellar bone. It is interesting that the bone tissue was undergoing remodeling, as evidenced by the presence of osteoclasts. Osteoclasts and odontoclasts can be observed in the same area and show morphologic similarity.

glass ionomer cement. Biodentine may prove to be a particularly suitable material for restoring these defects[28,80] because it may combine acceptable esthetics with the ability to support PDL attachment.

Once the cavity has been restored, the mucoperiosteal flap is replaced and secured in position. If perforation of the root canal wall has occurred, root canal treatment should be carried out. Access to the root canal system should be gained under a rubber dam before the resorptive defect is assessed. The root canal should be prepared in the area of the defect as normal, using saline as an irrigant. A tapered gutta-percha (GP) point then should be placed in the canal to maintain its patency during the excavation and restoration of the resorptive defect and to provide a barrier against which the final restoration can be condensed. The rubber dam is removed, and surgical treatment of the resorptive defect can be carried out as described, without any risk of the resorptive debris entering the root canal system. After repositioning of the mucoperiosteal flap, root canal treatment can be completed in the normal manner, without fear of extrusion of infected tissue, irrigants, or medicaments into the periodontal tissues.

One of the treatment modalities of ECR is internal repair of the lesion, which does not require any surgical procedure.[85] This treatment option is indicated when the resorptive defect is close to or has just perforated the pulp

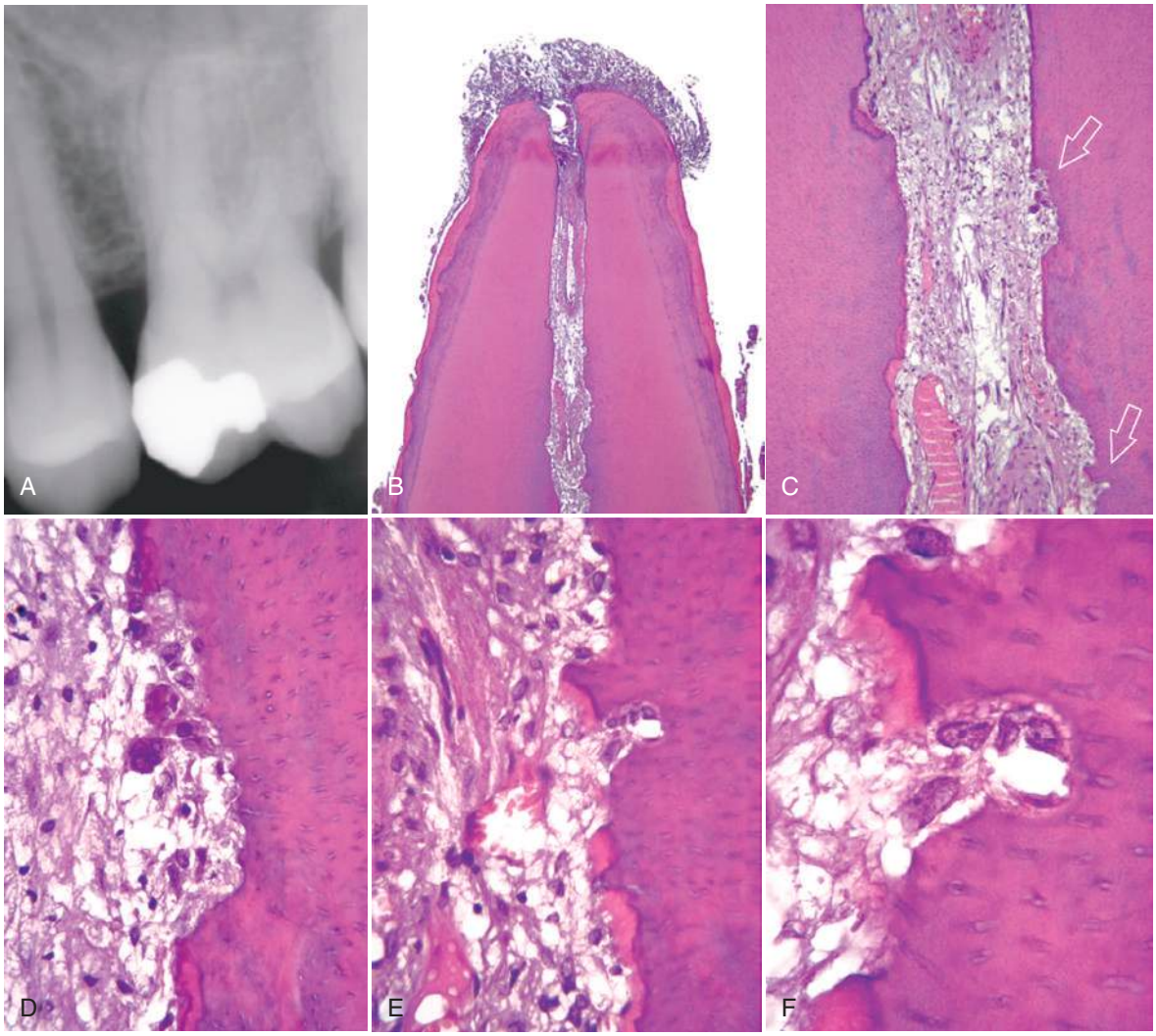



**Fig. 18.13** Early internal root resorption. **A,** A 54-year-old male patient presented with a long story of pain on chewing and with cold stimuli. Recently the pain had become continuous and severe. The maxillary first molar, which had a mesial amalgam restoration, did not respond to sensitivity tests. A radiograph showed a periapical radiolucency on the palatal root. After removal of the restorative materials, a crack line involving the pulp chamber floor was diagnosed, and the tooth was extracted. **B,** Some pathologic tissue remained attached to the palatal root apex at extraction. A longitudinal section passing at the center of the canal demonstrated vital connective tissue in the apical third. (H&E stain; original magnification ×16.) **C,** Detail of the canal in **B.** Note the connective tissue with vessels and relatively few inflammatory cells, in addition to the resorption lacunae on the root canal walls (original magnification ×1000). **D,** High-power view of the area of the canal wall indicated by the upper *arrow* in **C.** Note the resorption lacuna with odontoclasts (original magnification ×1000). **E** and **F,** Progressive magnification of the area of the canal wall indicated by the lower *arrow* in **C.** Note the Howship lacuna housing clastic cells. Predentin is present in some areas of the wall, but it is absent in the areas with active resorption (original magnification ×400 and ×1000).

**Table 18.1** Three-Dimensional Classification of External Cervical Resorption

| Hight | Circumferential spread | Proximity to the root canal |
|---|---|---|
| 1. At cemento-enamel junction level or coronal to the bone crest (supracrestal) | A: ≤90° | d: Lesion confined to dentine |
| 2. Extends into coronal third of the root and apical to the bone crest (subcrestal) | B: >90° to ≤180° | p: Probable pulpal involement |
| 3. Extends into mid-third of the root | C: >180° to ≤270° | |
| 4. Extends into apical third of the root | D: >270° | |

chamber, and external surgical flap is not accessible and/or result in excessive removal of sound tooth structure. The internal repair is preferred in cases where the entry point is small, intraosseous, and apical to the epithelial junction.

Root canal treatment in such cases might be indicated if the defect was close to the pulp. Endodontic treatment could be done in one visit. After completion of the endodontic treatment, the access cavity should be extended to encompass the ECR lesion. The ECR and endodontic access cavity may be restored with a direct plastic restoration such as composite resin or glass-ionomer cement. Alternatively, Biodentine could be used as a dentine substitute, over which a thin layer of glass ionomer cement and/or composite resin may be placed.[90,99]

The treatment of ECR depends on the severity, extent, and location of the resorptive defect and the restorability of the tooth. Heithersay[55] developed a four-stage classification system for ECR based on the depth of penetration of the resorption in a buccolingual and apicocoronal direction. He examined the prognosis of treatment in 101 cases of ECR in 94 patients using the nonsurgical protocol referred to previously and related the success rates for treatment to the classification of the lesion.[55] He reported a 100% success rate for class I and class II lesions, a 77.8% success rate for class III lesions, and a 12.5% success rate for class IV lesions. This emphasizes the poorer outcome that can be expected for more advanced cases. A major limitation of the Heithersay classification is that it is valid only if the ECR lesion is confined to the proximal aspect of the tooth, because lesions are assessed on 2D radiographs. If the ECR lesion is located on and/or extends to the labial and/or buccal (proximal) aspects of the tooth, the true nature of lesion cannot be accurately assessed with radiographs.[81,87]

# Internal Root Resorption

## INTRODUCTION

IRR is a form of root resorption that originates in and affects the root canal wall.[6] It is further classified as either inflammatory or replacement. The replacement type is associated with the deposition of mineralized tissue in the root canal space after the initial loss of dentin (initial dentin loss is a feature of both types).[6] Because their characteristics are largely similar, the two types of IRR are discussed together.

## ETIOLOGY AND PATHOGENESIS OF INTERNAL ROOT RESORPTION

Damage to the outermost protective odontoblast layer and the predentin of the canal wall results in exposure of the underlying mineralized dentin to odontoclasts, which subsequently leads to IRR (Fig. 18.14).[108,118]

The exact cause or type of injury that causes such damage is not completely understood. Several etiologic factors have been expressed for the loss of predentin, including trauma, caries and periodontal infections, excessive heat generated during restorative procedures on vital teeth, calcium hydroxide procedures, vital root resections, anachoresis, orthodontic treatment, cracked teeth, or simply idiopathic dystrophic changes in normal pulps.[7,17,26,92,115] In a study of 25 teeth with internal resorption, trauma was found to be the most common predisposing factor, responsible for 45% of the cases examined.[27] The suggested etiologies in the other cases were inflammation as a result of carious lesions (25%) and carious/periodontal lesions (14%). The cause of the internal resorption in the remaining teeth was unknown. Other reports in the literature support the view that trauma[7,119,121] and pulpal inflammation/infection[52,121] are the major contributory factors in the initiation of internal resorption.

As previously mentioned, damage to the odontoblast layer and predentin of the canal wall is a prerequisite for initiation of IRR.[119] However, progression of IRR depends on bacterial stimulation of the clastic cells involved in hard tissue resorption. Without this stimulation, the resorption is self-limiting.[119]

For IRR to continue, the pulp tissue apical to the resorptive lesion must have a viable blood supply; this provides clastic cells and their nutrients, and the infected necrotic coronal pulp tissue provides stimulation for those clastic cells (see Fig. 18.14).[107] Bacteria may enter the pulp canal through dentinal tubules, carious cavities, cracks, fractures, and lateral canals. In the absence of a bacterial stimulus, the resorption is transient and may not advance to the stage that can be diagnosed clinically and radiographically. Therefore, the pulp apical to the site of resorption must be vital for the resorptive lesion to progress (see Fig. 18.14). If left untreated, internal resorption may continue until the inflamed connective tissue filling the resorptive defect degenerates, advancing the lesion in an apical direction. Ultimately, if left untreated, the pulp tissue apical to the resorptive lesion undergoes necrosis and the bacteria infect the entire root canal system, resulting in apical periodontitis.[95]

## HISTOLOGIC APPEARANCE

Wedenberg and Zetterqvist[121] reported on the histologic nature of IRR. The authors specifically examined the histologic, enzyme histochemical, and scanning electron micrographic (SEM) features of the resorptive process in a small sample of primary and secondary teeth extracted due to progressive IRR. The histologic and enzyme histochemical profiles were identical for the two groups, but the resorptive process appeared to occur more rapidly in the primary teeth. The pulpal tissue was populated to varying degrees in all teeth with an inflammatory infiltrate composed predominantly of lymphocytes and macrophages, with some neutrophils. The connective tissue in the pulp spaces was less vascular than healthy pulpal tissue and resembled periodontal membrane connective tissue with relatively more cells and fibers. The odontoblast layer and predentin were absent from the affected dentinal walls, which were populated by large, multinucleated odontoclasts occupying resorption lacunae. The odontoclasts showed evidence of active resorption (see Fig. 18.14). They were accompanied by mononuclear cells, believed to be odontoclast precursors, which populated the connective tissue adjacent to the site of resorption. Both types of cell displayed tartrate-resistant acid phosphatase (TRAP) activity.

Interestingly, the root canal wall was incompletely lined with a mineralized tissue resembling bone or cementum in all of the examined teeth.[121] Furthermore, islands of

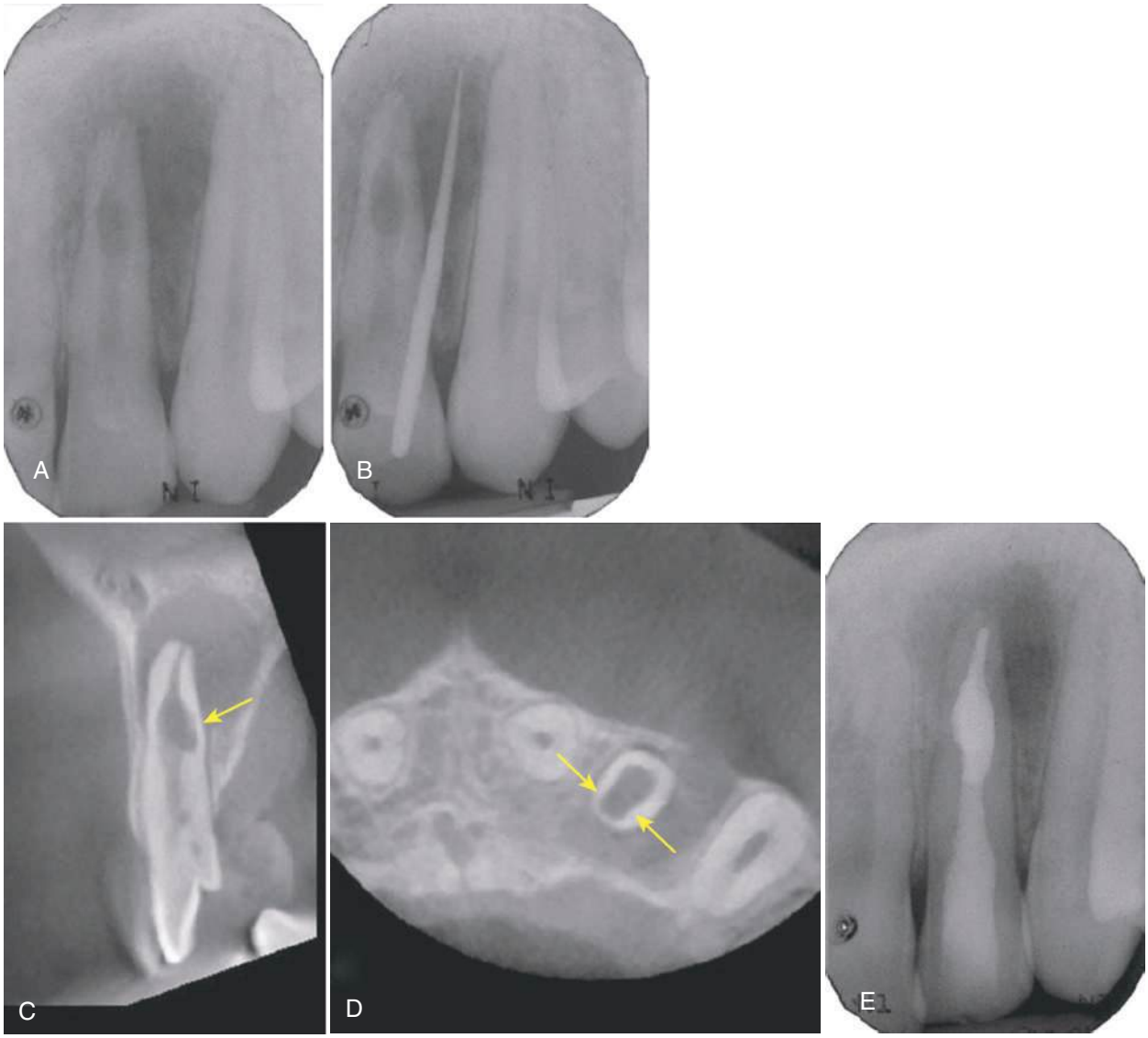



**Fig. 18.14** **A** and **B,** Parallax views of the maxillary left lateral incisor showing internal root resorption with necrosis. A gutta-percha point has been used to track the sinus. The reconstructed sagittal **(C)** and axial **(D)** slices from a CBCT scan reveal that the lesion has resorbed the palatal aspect of the root *(arrows)* and has nearly perforated the root wall. **E,** The tooth has been obturated with gutta-percha using a thermoplasticized technique. (From Patel S, Ricucci D, Durack C, Tay F: Internal root resorption: a review. *J Endod* 36:1107–1121, 2010.)

calcified tissue of a similar nature occupied the pulpal space in three of the cases. Islands of mineralized tissue occupying the root canal space are the defining feature of internal replacement resorption.[6] The authors suggested that deposition of this mineralized tissue is likely to be part of a coupling process, at the end of a period of resorption, in which osteoblasts are attracted to the affected site and participate in bone formation.[121]

## CLINICAL FEATURES

The clinical features of IRR largely depend on the histologic status of the affected pulp, the extent of the hard tissue destruction caused by the resorptive process, and the position of the resorptive cavity in the root canal space. In the active stages of resorption, bacterial contamination of vital pulpal tissue may cause an acute inflammatory response, leading to clinical symptoms of pulpitis. With the onset of pulpal necrosis and an established bacterial colonization of the root canal space, clinical signs and symptoms associated with acute or chronic apical periodontitis may develop. Sinus tract(s) may occur and may be associated with suppuration in the periapical tissues or possibly at the site of a perforation of the root canal wall caused by the hard tissue destruction. Extensive resorption of the coronal pulp may result in a pink or red discoloration visible through the crown of the affected tooth; this is caused by granulomatous tissue extending into and occupying the resorptive defect.[67] Although often reported as a common clinical indicator of the process, these pink spots actually are rare in cases of IRR. They may occur relatively more frequently in cases of ECR, but they are not very common with that resorption type, either. Often the affected tooth is asymptomatic, and clinical signs are absent.

## RADIOGRAPHIC FEATURES AND DIAGNOSIS

The diagnosis of any type of root resorption depends on radiographic demonstration of its presence. Two-dimensional radiographs make the detection and differentiation of the various types of resorption challenging. Both IRR and ECR could have similar features which makes differentiation and diagnosis from a two-dimensional shadowgraph difficult.[51,84,88,107] Gartner et al.[46] reported that lesions of IRR present radiographically as radiolucencies of uniform density that have a smooth outline and are symmetrically distributed over the root of the affected tooth (Fig. 18.15). The authors further reported that the outline of the root canal wall should not be traceable through the resorption defect because the root canal wall balloons out. Other authors have described IRR lesions as oval, circumscribed radiolucencies in continuity with the root canal wall.[77] Although certain cases of IRR may have some or all of these radiographic features, many do not; each case should be assessed individually before a diagnosis is made.

IRR can occur at any location in the root canal system and may manifest radiographically as a radiolucency with variable shape, radiodensity, outline, and symmetry in relation to the root canal. Internal inflammatory root resorption lesions are more likely be uniformly radiolucent, whereas in internal replacement (metaplastic) root resorption, the defect has a somewhat mottled or clouded appearance as a result of the radiopaque nature of the calcified material occupying the lesion (Fig. 18.16).[88] ECR lesions may contain predominantly granulomatous tissue, predominantly calcified tissue, or a mixture of the two; therefore, they may have a radiodensity similar to either type of internal resorption, which complicates the clinical differentiation of the disease process. As noted by Gartner et al.,[46] the best practice is to trace the outline of the root canal wall as it approaches and leaves the resorption defect. An IRR cavity is continuous with the normal root canal walls because it is essentially an extension of them. As such, in teeth with single canals affected by IRR, the canal walls should not be traceable through the defect. This is in contrast to ECR, in which the lesion lies buccal or palatal/lingual to the defect and is consequently superimposed on the canal system when viewed on conventional radiographs. In this situation the canal walls should maintain their normal course as they pass through the resorption defect, allowing them to be traced through it. However, it should be noted that in teeth with multiple canals, a canal that has been unaffected by IRR may be superimposed onto the IRR resorption defect on conventional radiographs. When the guidelines outlined by Gartner et al.[46] are used, this may lead to misdiagnosis

Parallax radiographs must always be used to obtain further information about the resorptive process. In addition to a paralleled periapical radiograph, a radiograph should be taken with a shift in the horizontal angulation of the x-ray tube in relation to the image receptor. IRR lesions maintain their position relative to the root canal system on the angled view. ECR lesions move in the same direction as the x-ray tube shift if they are lingually/palatally positioned and move in the opposite direction if they are buccally located.[46] This diagnostic technique, coupled with tracing of the root canal/pulp chamber outline through the lesion, have been the most reliable aids in the differential diagnosis of IRR when conventional radiography is used. However, as discussed previously, the amount of information available from conventional radiographic imaging is limited. This can lead to misdiagnoses and incorrect treatment in the management of IRR and invasive cervical resorption.

The use of CBCT as a diagnostic and treatment planning tool in the management of IRR has been reported in the literature.[20] Information such as the position, extent, and dimensions of an IRR lesion, in addition to the presence of any associated perforation, can be obtained from a CBCT scan. The same scan can differentiate between ECR and IRR, removing any doubt about the diagnosis that may have arisen with the conventional radiographic examination.

## MANAGEMENT

Once a diagnosis of IRR has been made, the extent of the hard tissue destruction must be assessed, and a clinical decision must be made about the prognosis of the affected tooth. If the affected tooth is salvageable and has a reasonable prognosis, root canal treatment is necessary. As with any infected tooth, the main purpose of the root canal treatment is to remove the intraradicular bacteria and disinfect the root canal space. If the resorptive process is still active, the treatment serves an adjunctive purpose, which is to eliminate the vital apical tissue that is sustaining and stimulating the resorbing cells.

The nature of the resorptive process in cases of IRR presents the endodontist with unique operative challenges. In teeth with active resorption, profuse bleeding from the granulomatous and inflamed pulpal tissues may impair visibility in the initial stages of treatment and may provide a stubborn source of mild hemorrhage when attempts are made to dry the canal after chemomechanical preparation. Furthermore, the irregularly concave nature of resorption defects makes them inaccessible to direct mechanical debridement.

## CHEMOMECHANICAL DEBRIDEMENT OF THE ROOT CANAL

Teeth usually have unique morphologic features of root canal that harbor microorganisms in infected teeth. Endodontic instruments and passively delivered irrigants fail to penetrate into these secluded spaces and niches.[75,96,101] The use of ultrasonic instruments to aid the penetration of endodontic irrigants has been shown to improve the removal of organic debris and biofilms from the root canal space.[24] Given the inaccessibility of IRR defects to normal instrumentation and passive irrigation, ultrasonic activation of irrigants should be considered an essential step in the treatment of these cases (see Fig. 18.14). However, even when this adjunctive measure is used, microbes may persist in confined areas after chemomechanical debridement.[24] As such, an intracanal antibacterial medicament should be used to further reduce the microbial load and improve the disinfection of the root canal space.[101] Calcium hydroxide is an antibacterial, interappointment, endodontic medicament that has been shown to eradicate bacteria persisting in the root canal space after root canal treatment.[25,102]

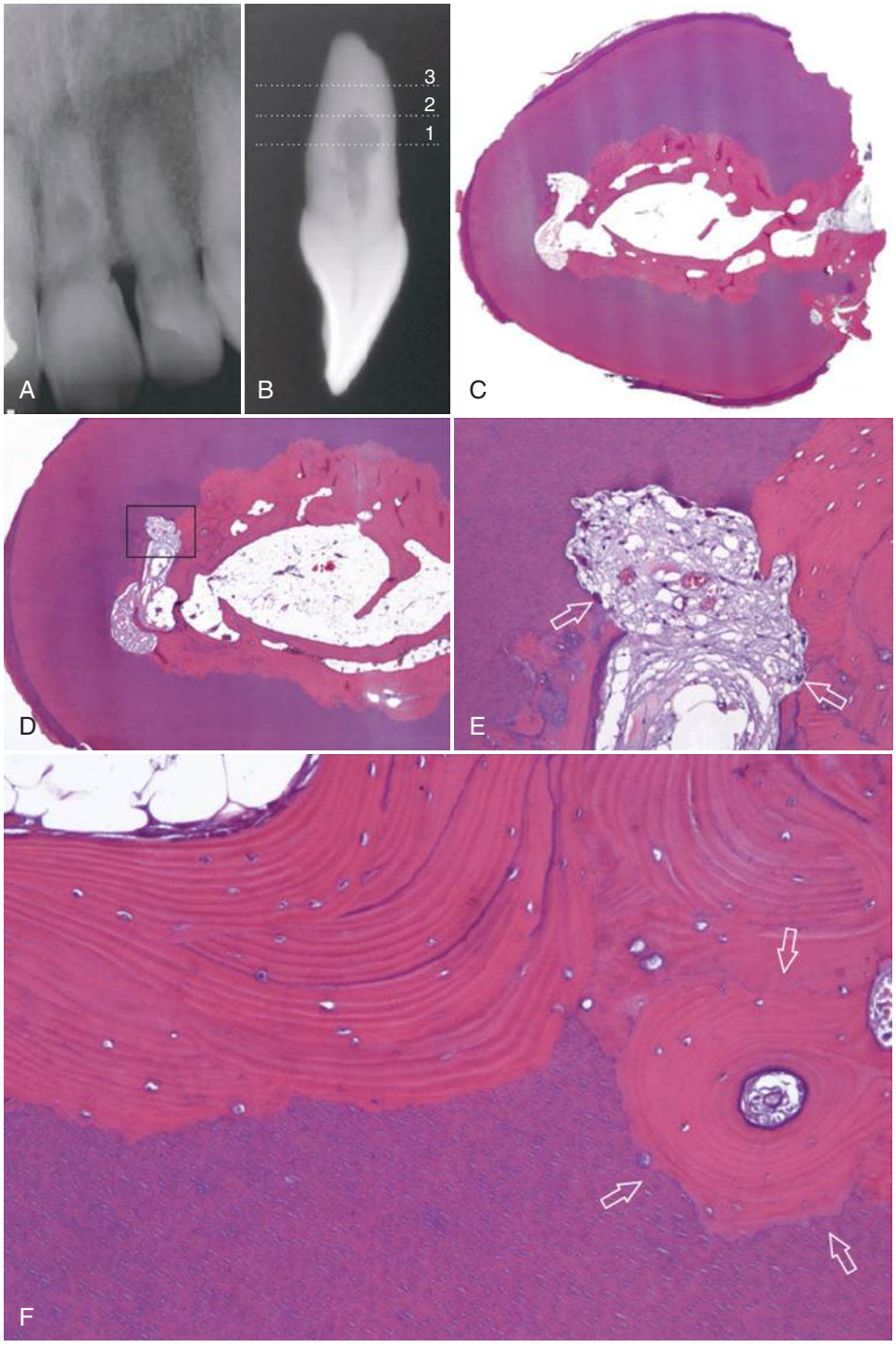



**Fig. 18.15** Light microscopy images of a tooth with internal (root canal) replacement resorption. The tooth belonged to a 44-year-old male patient who was referred to the first author for management of a perforated root. The tooth was asymptomatic on examination, but there was a history of previous trauma. **A,** Radiograph of a maxillary central incisor with a radiolucent lesion in the middle third of the root canal. The radiolucent lesion appears to be mottled, which suggests internal root resorption with metaplasia. **B,** Clinical radiograph of the tooth after extraction, taken at a 90-degree angle, showing the continuity of the resorptive lesion with the canal space. **C,** Cross section taken approximately at the level of *line 1* in *B.* The low-magnification overview shows that the dentin around the root canal had been replaced by an ingrowth of bone tissue, and the root appears to have been perforated on the distopalatal aspect. (H&E stain; ×8.) **D,** Higher magnification of *C.* (H&E stain; ×16.) **E,** High magnification of the area demarcated by the rectangle in *D.* The intraradicular dentin has been resorbed. (H&E stain; ×100.) **F,** High-magnification view taken from the right part of *C,* showing that the resorbed dentin has been replaced by lamellar bone. Osteocytes are present in lacunae between the lamellae. A characteristic cross section of an osteon can be seen on the right *(open arrows),* with concentric lamellae surrounding a vascular structure. (H&E stain; ×100.)

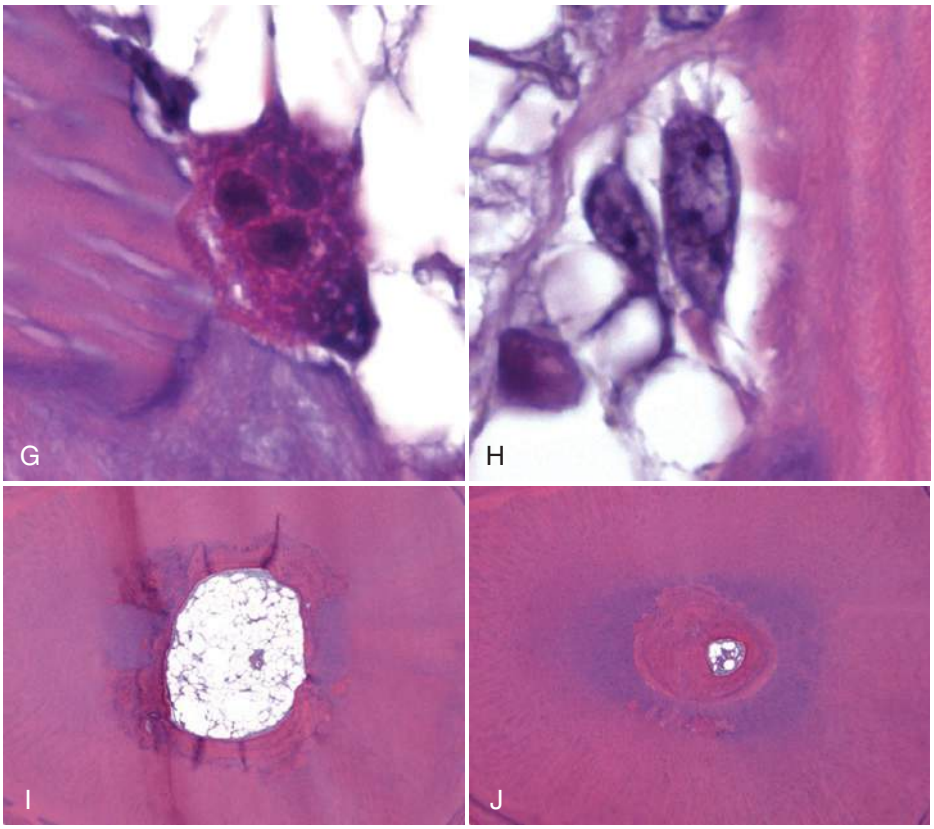



**Fig. 18.15, cont'd** **G,** High-magnification view of the area indicated by the left open *arrow* in *E.* A multinucleated resorbing cell (odontoclast) can be seen in a dentinal lacuna, indicating active resorption of the dentinal wall. (H&E stain; ×1000.) **H,** High-magnification view of the bone surface indicated by the right *arrow* in *E.* The large cells are osteoblast-like cells. Once they produced mineralized tissue, they were embedded in the bone lacunae, assuming the characteristics of osteocytes. (H&E stain; ×1000.) **I,** Cross section taken approximately at the level of *line 2* in *B.* The root canal is still large at this level and is surrounded by a relatively thin layer of newly formed bone. (H&E stain; ×16.) **J,** Cross section taken approximately at the level of *line 3* in *B.* At this level the root canal appears consistently narrowed by a dense layer of newly formed bone. (H&E stain; ×16.) (From Patel S, Ricucci D, Durack C, Tay F: Internal root resorption: a review. *J Endod* 36:1107–1121, 2010.)

Also, when used in conjunction with sodium hypochlorite, it potentiates the effect of that irrigant in the removal of organic debris from the root canal system.[1,110] Based on this evidence the authors advocate the use of calcium hydroxide as an intracanal, antibacterial medicament to supplement the conventional chemomechanical debridement of the root canal system.[88]

## OBTURATION

One of the fundamental objectives of endodontic treatment is to fill the disinfected root canal space completely with an appropriate material. In IRR, the hard tissue defects caused by the resorptive process are challenging to fill adequately. To obturate the resorptive defect, the root-filling material must be able to flow. GP is widely regarded as the gold standard filling material in endodontics. It can plasticize when pressure is applied, and it becomes flowable with the application of heat. Gencoglu et al.[47] examined the ability of different obturation systems and techniques to fill the defects in artificially created internal resorptive cavities ex vivo. They found that Obtura II (ObII) and Microseal (MS) thermoplastic GP systems produced significantly better fills in simulated resorptive cavities than did Thermafill, soft core systems (SCS), and cold lateral condensation (CLC). In a similar study Goldberg et al.[49] demonstrated that the Obtura II system filled simulated resorptive defects statistically better than CLC, Thermafill, and a hybrid technique. Gencoglu et al.[47] reported that the Obtura II and MS systems produced fills with greater GP–to-sealer (GP/sealer) ratios than did the other techniques examined. These findings were corroborated in the study by Goldberg et al.[49]

Because root canal sealers shrink on setting[122] and dissolve and degrade to varying degrees in the presence of moisture,[73] fillings with higher GP/sealer ratios reduce the risk of void formation and leakage of contaminants into the root canal system, with potentially positive benefits for the treatment outcome.

Stamos and Stamos[105] and Wilson and Barnes[123] reported on cases of internal resorption in which the Obtura system was successfully used to obturate the canal. In light of the evidence presented, the Obtura and MS systems apparently can be expected to produce the best technical results for obturating canals with IRR.

When choosing the appropriate materials and methods for filling resorptive defects of IRR, the clinician first must establish the presence and extent of any perforations in the wall of the affected root. This information can be readily obtained from an appropriate CBCT scan. If perforation has occurred, mineral trioxide aggregate (MTA) should be

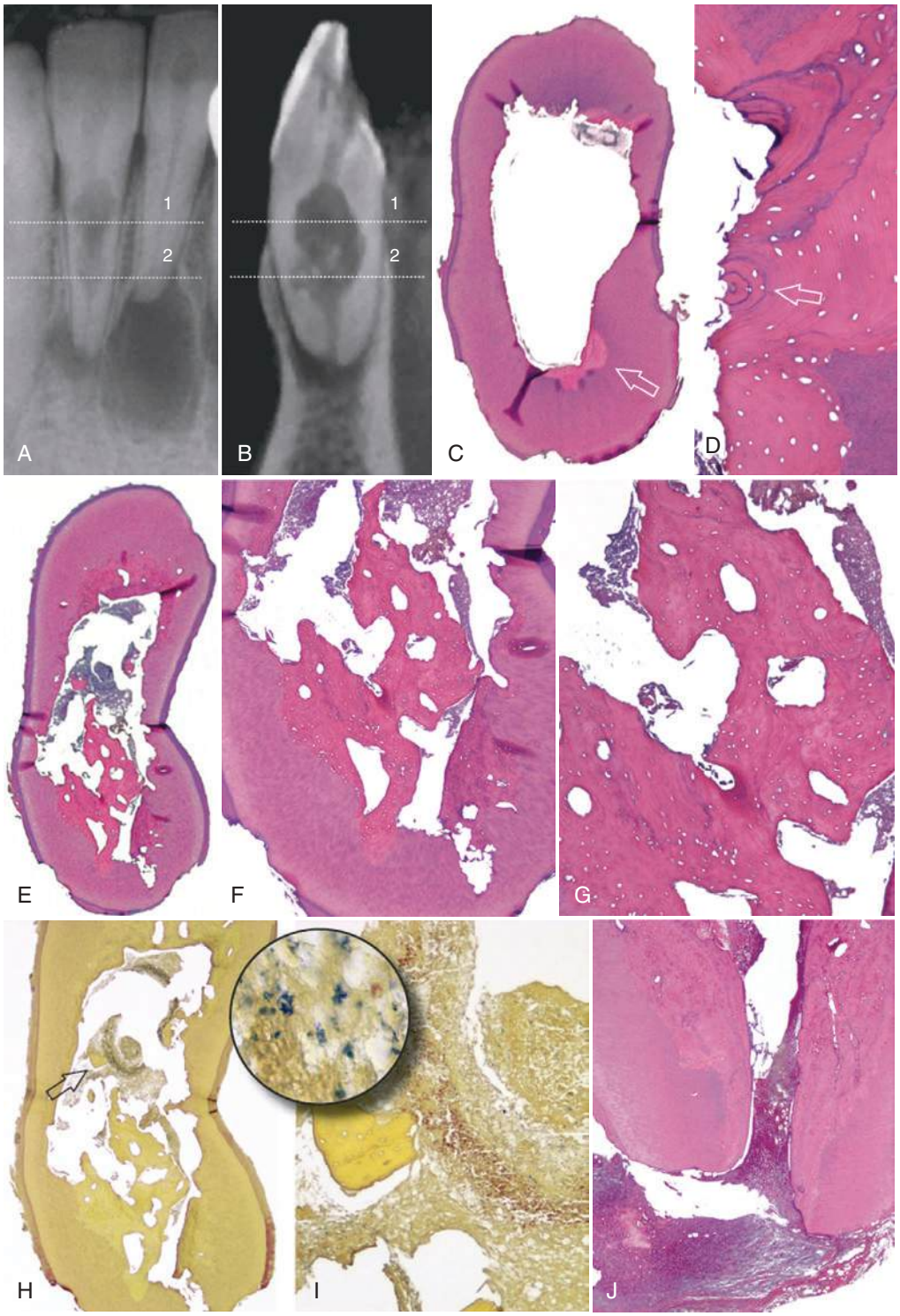



**Fig. 18.16** Light microscopy images of a variant of internal (root canal) replacement resorption with tunneling resorption. The lower right lateral incisor belonged to a 39-year-old former boxer who had suffered a jaw fracture in a boxing match in his early twenties and was placed in intermaxillary fixation. The patient developed symptoms 20 years later and complained of pain associated with his lower incisors. **A,** Radiograph of the mandibular right incisors. The lower right central incisor had asymptomatic apical periodontitis associated with a necrotic and infected pulp. The lower right lateral incisor showed a large area of internal root resorption. The tooth did not respond to sensitivity tests. **B,** Sagittal CBCT slice shows some calcified tissue in the resorptive defect. **C,** Cross section taken at the level of *line 1* in *A* and *B*. The overview shows that the canal was apparently empty at this level. (H&E stain; ×6.) **D,** High magnification of the area indicated by the *arrow* in *C*. Lamellar bone filling an area of previous resorption. Note the osteon structure *(arrow)*. (H&E stain; ×100.) **E,** Cross section taken at the level of line 2 in *A* and *B*. Overview shows that the canal lumen was partly occupied by necrotic remnants, partly by bonelike tissue. (H&E stain; ×8.) **F,** High magnification of the lower part in *E*. (H&E stain; ×50.) **G,** Higher magnification of *F*. Bone trabeculae are surrounded by necrotic debris. (H&E stain; ×100.) **H,** Cross section taken from the same area as that in *E*. (Taylor's modified Brown & Brenn [TBB] stain; ×16.) **I,** High magnification of the area indicated by the *arrow* in *H*. A fragment of bonelike tissue can be seen surrounded by bacteria-colonized necrotic tissues. (TBB stain; ×100; inset ×1000.) **J,** Longitudinal section passing approximately through the center of the root apex. Dentin walls have been resorbed and replaced by a bonelike tissue. (H&E stain; ×16.). (From Patel S, Ricucci D, Durack C, Tay F: Internal root resorption: a review. *J Endod* 36:1107–1121, 2010.)

considered the material of choice to repair the root wall. MTA is biocompatible,[74] has superior sealing properties when used as a retrograde filling material,[16] and has proved effective in the repair of lateral and furcal root perforations in animal studies.[68] Furthermore, the material is well tolerated in the periapical tissues, and when used as a root-end filling material in the absence of infection, it supports almost complete regeneration of the adjacent periodontium.[93] These are desirable properties in the context of perforation repair because of the very real possibility of unintentional extrusion of the material when a perforating internal resorptive defect is repaired in an orthograde manner. However, the flow properties of MTA are significantly poorer than those of heated GP. Its use as an effective filling material in IRR depends on adequate ultrasonic activation of the material to disperse it into the recesses of the defect.[20] Use of a dental operating microscope and the correct equipment to deliver the material are essential.

A hybrid technique to obturate canals affected by perforating internal resorption also may be used. In these cases, the canal apical to the resorption defect is filled with GP. The GP can then be used as a barrier against which the MTA can be packed. Hsiang-Chi Hsien et al.[57] successfully used this technique to treat a perforating IRR defect in a maxillary incisor. Jacobovitz and De Lima[59] described a case in which a maxillary central incisor with a large, perforating internal resorption defect had a poor prognosis and initially was designated for extraction. The tooth subsequently was successfully treated in an orthograde manner with white MTA and GP.[59]

Clinical situations arise in which a perforating resorptive defect causes extensive dental hard tissue destruction that fails to respond to or is not amenable to repair with an orthograde approach. Surgical treatment may be needed in these cases. For the reasons discussed already, MTA would be the material of choice to repair these perforations. In cases that have not already been treated in a nonsurgical manner, the operator first must access the root canal as for orthograde treatment. A well-fitting, tapered GP point or an appropriately sized finger spreader is then positioned in the canal to occlude it and to provide a barrier against which the MTA can be packed once surgical access to the defect has been gained. The barrier also prevents inadvertent deposition of the MTA into the apical third of the canal. The perforation is then exposed surgically and repaired with the MTA. The canal can be shaped, disinfected, and obturated with thermoplasticized GP once the MTA has set.

If the resorptive process has caused sufficient tissue destruction to render the tooth unrestorable, extraction is the

**Table 18.2 Characteristic Key Features of Root Resorption**

| | INTERNAL | | EXTERNAL | | | |
|---|---|---|---|---|---|---|
| **Resorption Type** | **Inflammatory** | **Replacement** | **Inflammatory** | **Replacement** | **Cervical** | **Surface** |
| Clinical finding | Discoloration, symptoms of irreversible pulpitis and/or apical periodontitis | Discoloration, Symptoms of irreversible pulpitis and/or apical periodontitis | Discoloration, symptoms of apical periodontitis | May exhibit a metallic percussion sound. Mobility not expected | Probable periodontal defect, symptoms of (ir)reversible pulpitis | None |
| Clinical appearance | Pink spot (rare), or sign of pulpitis/ apical periodontitis. | Pink spot (rare), or sign of pulpitis/ apical periodontitis. | Healthy, or sign of pulpitis/apical periodontitis. | Healthy | ±pink spot, or sign of pulpitis/apical periodontitis. | Healthy |
| Location on root | Anywhere | Anywhere | Anywhere | Anywhere | Cervical third initially but can be seen in mid and apical third in advanced cases | Adjacent to impacted tooth, cyst or tumour. Apical in orthodontically treated teeth |
| Pulp sensitivity | Maybe +ve in partially vital cases, or −ve in advance necrotic cases | Maybe +ve in partially vital cases, or −ve in advance necrotic cases | −ve | +ve | Usually +ve, –ve in advanced necrotic cases | +ve |
| Radiological appearance | Symmetrical oval shaped (ballooning) enlargement of root canal | Oval shaped enlargement of root canal with cloudy and/or mottled canal wall | Asymmetrical bowl shaped periradicular or periapical radiolucencies | Asymmetrical bony replacement of root surface and loss of PDL space | (A)symmetrical radiolucent (early), or radiopaque (advanced) appearance. Lesion moves on parallax views | Blunted root apex. Asymmetrical loss of root |
| Root canal | Canal expands into lesions | Canal expands into lesions | Intact in early EIR, perforation in advanced cases | Intact | Intact, but perforation in advanced cases | Intact |

*EIR*, External inflammatory resorption; *PDL*, periodontal ligament.
From Patel S, Saberi N: The ins and outs of root resorption. Br Dent J 224:691–699, 2018.[89]

most appropriate treatment option. If the tooth has been weakened by the disease process to the extent that fracture of the tooth root is likely, the patient may elect to have the tooth extracted. The presence of a perforating resorptive defect is certainly not a contraindication to treatment, but a perforation of significant size will have a bearing on the decision to surgically treat or extract the tooth. CBCT is an invaluable component of the clinician's armamentarium in the treatment of IRR.[38] A CBCT scan gives the clinician a 3D view of the tooth, the resorptive defect, and the adjacent anatomy. The clinician thus has the information necessary to determine a prognosis for the tooth and/or its amenability to surgical repair. If extraction of the affected tooth is indicated, the scan may be used as a diagnostic and treatment planning tool for provision of a dental implant–retained prosthesis. Bhuva et al.[20] described a case in which CBCT was used as a treatment planning tool in the successful treatment of a case of IRR. As mentioned previously, CBCT has been found to be highly accurate in revealing the nature of the resorptive lesion, and this leads to the selection of the most suitable treatment plan.

## Summary

- More research is needed for understanding of the etiology and pathogenesis of different types of root resorption (Table 18.2).
- New ECR classification provides a better description of the lesion in a three-dimensional manner.
- The earlier the detection of root resorption, the better the prognosis and the outcome.
- CBCT is an excellent diagnostic tool for the confirmation of the presence of ECR and IRR and also in appreciation of the true nature of these conditions.

## References


1. Andersen M, Lund A, Andreasen JO, et al: In vitro solubility of human pulp tissue in calcium hydroxide and sodium hypochlorite, *Endod Dent Traumatol* 8(3):104–108, 1992.
2. Andreasen FM, Pedersen BV: Prognosis of luxated permanent teeth—the development of pulp necrosis, *Endod Dent Traumatol* 1(6):207–220, 1985.
3. Andreasen FM, Sewerin I, Mandel U, et al: Radiographic assessment of simulated root resorption cavities, *Endod Dent Traumatol* 3(1):21–27, 1987.
4. Andreasen JO: Review of root resorption systems and models. Etiology of root resorption and the homeostatic mechanisms of the periodontal ligament. In Davidovitch Z, editor: *The biological mechanisms of tooth eruption and root resorption*, Birmingham 1988, EBSCO Media, pp 9–21.
5. Andreasen J: Avulsions. In Andreasen JO, Andreasen FM, Andersson L, editor: *Textbook and color atlas of traumatic injuries to the teeth*, ed 4, Copenhagen, 2007, Blackwell Munksgaard.
6. Andreasen J, Andreasen F: Root resorption following traumatic dental injuries. Proceedings of the Finnish Dental Society, *Suomen Hammaslaakariseuran Toimituksia*, 88:95–114, 1992.
7. Andreasen JO: Luxation of permanent teeth due to trauma. A clinical and radiographic follow-up study of 189 injured teeth, *Scand J Dent Res* 78(3):273–286, 1970.
8. Andreasen JO: A time-related study of periodontal healing and root resorption activity after replantation of mature permanent incisors in monkeys, *Swed Dent J* 4(3):101–110, 1980.
9. Andreasen JO: Relationship between surface and inflammatory resorption and changes in the pulp after replantation of permanent incisors in monkeys, *J Endod* 7(7):294–301, 1981.
10. Andreasen JO, Borum MK, Andreasen FM: Replantation of 400 avulsed permanent incisors. 3. Factors related to root growth, *Endod Dent Traumatol* 11(2):69–75, 1995.
11. Andreasen JO, Borum MK, Jacobsen HL, et al: Replantation of 400 avulsed permanent incisors. 2. Factors related to pulpal healing, *Endod Dent Traumatol* 11(2):59–68, 1995.
12. Andreasen JO, Farik B, Munksgaard EC: Long-term calcium hydroxide as a root canal dressing may increase risk of root fracture, *Dent Traumatol* 18(3):134–137, 2002.
13. Andreasen JO, Hjorting-Hansen E: Replantation of teeth. I. Radiographic and clinical study of 110 human teeth replanted after accidental loss, *Acta Odontol Scand* 24(3):263–286, 1966.
14. Andreasen JO, Hjorting-Hansen E: Replantation of teeth. II. Histological study of 22 replanted anterior teeth in humans, *Acta Odontol Scand* 24(3):287–306, 1966.
15. Andreasen JO, Lovschall H: Response of oral tissues to trauma. In *Traumatic injuries to the teeth*, Oxford, 2007, Blackwell Pub. Professional, pp 62–113.
16. Aqrabawi J: Sealing ability of amalgam, super EBA cement, and MTA when used as retrograde filling materials, *Br Dent J* 188(5):266–268, 2000.
17. Ashrafi MH, Sadeghi EM: Idiopathic multiple internal resorption: report of case, *ASDC J Dent Child* 47(3):196–199, 1980.
18. Bell T: The anatomy, physiology, and diseases of the teeth. *The North American Medical and Surgical Journal (1826-1831)*, 10(19):63, 1830, ARTICLE V.
19. Bender IB, Seltzer S: Roentgenographic and direct observation of experimental lesions in bone: I. 1961, *J Endod*, 29(11):702–706; discussion 701, 2003.
20. Bhuva B, Barnes JJ, Patel S: The use of limited cone beam computed tomography in the diagnosis and management of a case of perforating internal root resorption, *Int Endod J* 44(8):777–786, 2011.
21. Bille ML, Kvetny MJ, Kjaer I: A possible association between early apical resorption of primary teeth and ectodermal characteristics of the permanent dentition, *Eur J Orthod* 30(4):346–351, 2008.
22. Bille ML, Nolting D, Kvetny MJ, et al: Unexpected early apical resorption of primary molars and canines, *Eur Arch Paediatr Dent* 8(3):144–149, 2007.
23. Boyce BF, Xing L: Functions of RANKL/RANK/OPG in bone modeling and remodeling, *Arch Biochem Biophys* 473(2):139–146, 2008.
24. Burleson A, Nusstein J, Reader A, et al: The in vivo evaluation of hand/rotary/ultrasound instrumentation in necrotic, human mandibular molars, *J Endod* 33(7):782–787, 2007.
25. Bystrom A, Claesson R, Sundqvist G: The antibacterial effect of camphorated paramonochlorophenol, camphorated phenol and calcium hydroxide in the treatment of infected root canals, *Endod Dent Traumatol* 1(5):170–175, 1985.
26. Cabrini RL, Manfredi EE: Internal resorption of dentine; histopathologic control of eight cases after pulp amputation and capping with calcium hydroxide, *Oral Surg Oral Med Oral Pathol* 10(1):90–96, 1957.
27. Caliskan MK, Turkun M: Prognosis of permanent teeth with internal resorption: a clinical review, *Endod Dent Traumatol* 13(2):75–81, 1997.
28. Camilleri J, Investigation of Biodentine as dentine replacement material, *J Dent* 41(7):600–610, 2013.
29. Chow J, Chambers TJ: An assessment of the prevalence of organic material on bone surfaces, *Calcif Tissue Int* 50(2):118–122, 1992.
30. Chung CJ, Soma K, Rittling SR, et al: OPN deficiency suppresses appearance of odontoclastic cells and resorption of the tooth root induced by experimental force application, *J Cell Physiol* 214(3):614–620, 2008.
31. Cohenca N, Simon JH, Mathur A, et al: Clinical indications for digital imaging in dento-alveolar trauma. Part 2: root resorption, *Dent Traumatol* 23(2):105–113, 2007.
32. Crona-Larsson G, Bjarnason S, Noren JG: Effect of luxation injuries on permanent teeth, *Endod Dent Traumatol* 7(5):199–206, 1991.
33. Cvek M, Treatment of non-vital peranent incisors with calcium hydroxide, II. Effect on external root resorption in luxated teeth compared with effect of root filling with guttapercha. A foolow-up, *Odont Revy*, 24:343–354, 1973.

34. Cvek M, Prognosis of luxated non-vital maxillary incisors treated with calcium hydroxide and filled with gutta-percha. A retrospective clinical study, *Endod Dent Traumatol* 8(2):45–55, 1992.
35. Cvek M, Endodontic management and the use of calcium hydroxide in traumatized permanent teeth. In *Textbook and color atlas of traumatic injuries to the teeth*. ed 4, Ames, IA, 2007, Blackwell Munksgaard, pp 598–657.
36. D'Addazio PS, Campos CN, Özcan M, et al: A comparative study between cone-beam computed tomography and periapical radiographs in the diagnosis of simulated endodontic complications, *Int Endod J* 44(3):218–224, 2011.
37. Dumsha T, Hovland EJ: Evaluation of long-term calcium hydroxide treatment in avulsed teeth—an in vivo study, *Int Endod J* 28(1):7–11, 1995.
38. Durack C, Patel S: Cone beam computed tomography in endodontics, *Braz Dent J* 23(3):179–191, 2012.
39. Ehnevid H, Lindskog S, Jansson L, et al: Tissue formation on cementum surfaces in vivo, *Swed Dent J* 17(1-2):1–8, 1993.
40. Estrela C, Bueno MR, De Alencar AH, et al: Method to evaluate inflammatory root resorption by using cone beam computed tomography, *J Endod* 35(11):1491–1497, 2009.
41. Diangelis AJ, Andreasen JO, Ebeleseder KA, et al: Guidelines for the management of traumatic dental injuries. I. Fractures and luxations of permanent teeth, *Dent Traumatol* 23(2):66–71, 2007.
42. Flores MT, Andersson L, Andreasen JO, et al: Guidelines for the management of traumatic dental injuries. II. Avulsion of permanent teeth, *Dent Traumatol* 23(3):130–136, 2007.
43. Frank AL, Bakland LK: Nonendodontic therapy for supraosseous extracanal invasive resorption, *J Endod* 13(7):348–355, 1987.
44. Furseth R: The resorption processes of human deciduous teeth studied by light microscopy, microradiography and electron microscopy, *Arch Oral Biol* 13(4):417–431, 1968.
45. Fuss Z, Tsesis I, Lin S: Root resorption—diagnosis, classification and treatment choices based on stimulation factors, *Dent Traumatol* 19(4):175–182, 2003.
46. Gartner AH, Mack T, Somerlott RG, et al: Differential diagnosis of internal and external root resorption, *J Endod* 2(11):329–334, 1976.
47. Gencoglu N, Yildirim T, Garip Y, et al: Effectiveness of different gutta-percha techniques when filling experimental internal resorptive cavities, *Int Endod J* 41(10):836–842, 2008.
48. Gold SI, Hasselgren G: Peripheral inflammatory root resorption: a review of the literature with case reports, *J Clin Periodontol* 19(8):523–534, 1992.
49. Goldberg F, Massone EJ, Esmoris M, et al: Comparison of different techniques for obturating experimental internal resorptive cavities, *Dent Traumatol* 16(3):116–121, 2000.
50. Gröndahl HG, Huumonen S: Radiographic manifestations of periapical inflammatory lesions: how new radiological techniques may improve endodontic diagnosis and treatment planning, *Endod Topics* 8(1):55–67, 2004.
51. Gulabivala K, Searson LJ: Clinical diagnosis of internal resorption: an exception to the rule, *Int Endod J* 28(5):255–260, 1995.
52. Haapasalo M, Endal U: Internal inflammatory root resorption: the unknown resorption of the tooth, *Endod Topics* 14(1):60–79, 2006.
53. Hammarström L, Lindskog S: General morphological aspects of resorption of teeth and alveolar bone, *Int Endod J* 18(2):93–108, 1985.
54. Harokopakis-Hajishengallis E: Physiologic root resorption in primary teeth: molecular and histological events, *J Oral Sci* 49(1):1–12, 2007.
55. Heithersay GS: Invasive cervical resorption: an analysis of potential predisposing factors, *Quintessence Int* 30(2):83–95, 1999.
56. Heithersay GS: Invasive cervical resorption, *Endod Topics* 7(1):73–92, 2004.
57. Hsien HC, Cheng YA, Lee YL, et al: Repair of perforating internal resorption with mineral trioxide aggregate: a case report, *J Endod* 29(8):538–539, 2003.
58. Ishijima M, Rittling SR, Yamashita T, et al: Enhancement of osteoclastic bone resorption and suppression of osteoblastic bone formation in response to reduced mechanical stress do not occur in the absence of osteopontin, *J Exp Med* 193(3):399–404, 2001.
59. Jacobovitz M, De Lima R: Treatment of inflammatory internal root resorption with mineral trioxide aggregate: a case report, *Int Endod J* 41(10):905–912, 2008.
60. Jones SJ Boyd A, The resorption of dentine and cementum in vivo and in vitro. In Davidotch Z, editor: *Biological mechanisms of tooth eruption and root resorption*, Birmingham, AL, 1988, EBSCO Media.
61. Kundel HL, Revesz G: Lesion conspicuity, structured noise, and film reader error, *AJR Am J Roentgenol* 126(6):1233–1238, 1976.
62. Laux M, Abbott PV, Pajarola G, et al: Apical inflammatory root resorption: a correlative radiographic and histological assessment, *Int Endod J* 33(6):483–493, 2000.
63. Levander E, Malmgren O: Evaluation of the risk of root resorption during orthodontic treatment: a study of upper incisors, *Eur J Orthod* 10(1):30–38, 1988.
64. Levin L, Trope M: Root resorption. In Hargreaves KM, Goodis HE, editors: *Seltzer and Bender's dental pulp,* Chicago, IL, 2002, Quintessence Publishing, pp 425–448.
65. Li B, Long X, Wang S, et al: Clinical and radiologic features of desmoplastic ameloblastoma, *J Oral Maxillofac Surg* 69(8):2173–2185, 2011.
66. Lindskog S, Blomlof L, Hammarstrom L: Repair of periodontal tissues in vivo and in vitro, *J Clin Periodontol* 10(2):188–205, 1983.
67. Lyroudia KM, Dourou VI, Pantelidou OC, et al: Internal root resorption studied by radiography, stereomicroscope, scanning electron microscope and computerized 3D reconstructive method, *Dent Traumatol* 18(3):148–152, 2002.
68. Main C, Mirzayan N, Shabahang S, et al: Repair of root perforations using mineral trioxide aggregate: a long-term study, *J Endod* 30(2):80–83, 2004.
69. Mavridou AM, Bergmans L, Barendregt D, et al: Descriptive analysis of factors associated with external cervical resorption, *J Endod* 43(10):1602–1610, 2017.
70. Mavridou AM, Hauben E, Wevers M, et al: Understanding external cervical resorption in vital teeth, *J Endod* 42(12):1737–1751, 2016.
71. Mavridou AM, Hilkens P, Lambrichts I, et al: Is hypoxia related to external cervical resorption? a case report, *J Endod* 45(4):459–470, 2019.
72. McHugh KP, Shen Z, Crotti TN, et al: Role of cell-matrix interactions in osteoclast differentiation, *Adv Exp Med Biol*, 602:107–111, 2007.
73. McMichen FR, Pearson G, Rahbaran S, et al: A comparative study of selected physical properties of five root-canal sealers, *Int Endod J* 36(9):629–635, 2003.
74. Mitchell PJ, Pitt Ford TR, Torabinejad M, et al: Osteoblast biocompatibility of mineral trioxide aggregate, *Biomaterials* 20(2):167–173, 1999.
75. Nair PN, Henry S, Cano V, et al: Microbial status of apical root canal system of human mandibular first molars with primary apical periodontitis after "one-visit" endodontic treatment, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 99(2):231–252, 2005.
76. Nakamura I, Duong LT, Rodan SB, et al: Involvement of $\alpha$ v $\beta$ 3 integrins in osteoclast function, *J Bone Miner Metab* 25(6):337–344, 2007.
77. Ne RF, Witherspoon DE, Gutmann JL: Tooth resorption, *Quintessence Int* 30(1):9–25, 1999.
78. Neuvald L, Consolaro A: Cementoenamel junction: microscopic analysis and external cervical resorption, *J Endod* 26(9):503–508, 2000.
79. Nilsen R, Magnusson BC: Enzyme histochemistry of induced heterotropic bone formation in guinea-pigs, *Arch Oral Biol*, 24(10-11):833–841, 1979.
80. Nowicka A, Lipski M, Parafiniuk M, et al: Response of human dental pulp capped with biodentine and mineral trioxide aggregate, *J Endod* 39(6):743–747, 2013.
81. Patel S, Dawood A: The use of cone beam computed tomography in the management of external cervical resorption lesions, *Int Endod J* 40(9):730–737, 2007.
82. Patel S, Dawood A, Whaites E, et al: New dimensions in endodontic imaging: part 1. Conventional and alternative radiographic systems, *Int Endod J* 42(6):447–462, 2009.
83. Patel S, Dawood A, Wilson R, et al: The detection and management of root resorption lesions using intraoral radiography and cone beam computed tomography - an in vivo investigation, *Int Endod J* 42(9):831–838, 2009.
84. Patel S, Ford TP: Is the resorption external or internal? *Dent Update*, 34(4):218–220, 222, 224-6, 229, 2007.
85. Patel S, Foschi F, Condon R, et al: External cervical resorption: part 2 - management, *Int Endod J* 51(11):1224–1238, 2018.
86. Patel S, Foschi F, Mannocci F, et al: External cervical resorption: a three-dimensional classification, *Int Endod J* 51(2):206–214, 2018.

87. Patel S, Kanagasingam S, Pitt Ford T: External cervical resorption: a review, *J Endod* 35(5):616–625, 2009.
88. Patel S, Ricucci D, Durak C, et al: Internal root resorption: a review, *J Endod* 36(7):1107–1121, 2010.
89. Patel S, Saberi N: The ins and outs of root resorption, *Br Dent J* 224(9):691–699, 2018.
90. Patel S, Vincer L: Case report: single visit indirect pulp cap using biodentine, *Dental Update* 44(2):141–145, 2017.
91. Pierce AM, Experimental basis for the management of dental resorption, *Endod Dent Traumatol* 5(6):255–265, 1989.
92. Rabinowitch BZ, Internal resorption, *Oral Surg Oral Med Oral Pathol* 33(2):263–282, 1972.
93. Regan JD, Gutmann JL, Witherspoon DE: Comparison of Diaket and MTA when used as root-end filling materials to support regeneration of the periradicular tissues, *Int Endod J* 35(10):840–847, 2002.
94. Revesz G, Kundel HL, Graber MA: The influence of structured noise on the detection of radiologic abnormalities, *Invest Radiol* 9(6):479–486, 1974.
95. Ricucci D: Apical limit of root canal instrumentation and obturation, part 1. Literature review, *Int Endod J* 31(6):384–393, 1998.
96. Ricucci D, Langeland K: Apical limit of root canal instrumentation and obturation, part 2. A histological study, *Int Endod J* 31(6):394–409, 1998.
97. Rodríguez G, Abella F, Durán-Sindreu F, et al: Influence of cone-beam computed tomography in clinical decision making among specialists, *J Endod* 43(2):194–199, 2017.
98. Rodríguez G, Patel S, Durán-Sindreu F, et al: Influence of cone-beam computed tomography on endodontic retreatment strategies among general dental practitioners and endodontists, *J Endod* 43(9):1433–1437, 2017.
99. Salzano S, Tirone F: Conservative nonsurgical treatment of class 4 invasive cervical resorption: a case series, *J Endod* 41(11):1907–1912, 2015.
100. Schaffner P, Dard M: Structure and function of RGD peptides involved in bone biology, *Cell Mol Life Sci* 60(1):119–132, 2003.
101. Siqueira JF Jr, Rôças IN, Santos SR, et al: Efficacy of instrumentation techniques and irrigation regimens in reducing the bacterial population within root canals, *J Endod* 28(3):181–184, 2002.
102. Sjögren U, Figdor D, Spångberg L, et al: The antimicrobial effect of calcium hydroxide as a short-term intracanal dressing, *Int Endod J* 24(3):119–125, 1991.
103. Soltanoff CS, Yang S, Chen W, et al: Signaling networks that control the lineage commitment and differentiation of bone cells, *Crit Rev Eukaryot Gene Expr* 19(1):1–46, 2009.
104. Speziani C, Rivollier A, Gallois A, et al: Murine dendritic cell transdifferentiation into osteoclasts is differentially regulated by innate and adaptive cytokines, *Eur J Immunol* 37(3):747–757, 2007.
105. Stamos DE, Stamos DG: A new treatment modality for internal resorption, *J Endod* 12(7):315–319, 1986.
106. Suei Y, Taguchi A, Nagasaki T, et al: Radiographic findings and prognosis of simple bone cysts of the jaws, *Dentomaxillofac Radiol* 39(2):65–71, 2010.
107. Tronstad L: Root resorption—etiology, terminology and clinical manifestations, *Endod Dent Traumatol* 4(6):241–252, 1988.
108. Trope M: Root resorption of dental and traumatic origin: classification based on etiology, *Pract Periodontics Aesthet Dent* 10(4):515–522, 1998.
109. Trope M: Root resorption due to dental trauma, *Endod Topics* 1(1):79–100, 2002.
110. Turkun M, Cengiz T: The effects of sodium hypochlorite and calcium hydroxide on tissue dissolution and root canal cleanliness, *Int Endod J* 30(5):335–342, 1997.
111. Tyrovola JB, Spyropoulos MN, Makou M, et al: Root resorption and the OPG/RANKL/RANK system: a mini review, *J Oral Sci* 50(4):367–376, 2008.
112. Uchiyama M, Nakamichi Y, Nakamura M, et al: Dental pulp and periodontal ligament cells support osteoclastic differentiation, *J Dent Res* 88(7):609–614, 2009.
113. Vaz de Souza D, Schirru E, Mannocci F, et al: External cervical resorption: a comparison of the diagnostic efficacy using 2 different cone-beam computed tomographic units and periapical radiographs, *J Endod* 43(1):121–125, 2017.
114. Wada N, Maeda H, Tanabe K, et al: Periodontal ligament cells secrete the factor that inhibits osteoclastic differentiation and function: the factor is osteoprotegerin/osteoclastogenesis inhibitory factor, *J Periodontal Res* 36(1):56–63, 2001.
115. Walton RE, Leonard LA: Cracked tooth: an etiology for "idiopathic" internal resorption? *J Endod* 12(4):167–169, 1986.
116. Webber RL, Messura JK: An in vivo comparison of diagnostic information obtained from tuned-aperture computed tomography and conventional dental radiographic imaging modalities, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 88(2):239–247, 1999.
117. Wedenberg C: Evidence for a dentin-derived inhibitor of macrophage spreading, *Scand J Dent Res* 95(5):381–388, 1987.
118. Wedenberg C, Lindskog S: Experimental internal resorption in monkey teeth, *Endod Dent Traumatol* 1(6):221–227, 1985.
119. Wedenberg C, Lindskog S: Evidence for a resorption inhibitor in dentin, *Scand J Dent Res* 95(3):205–211, 1987.
120. Wedenberg C, Yumita S: Evidence for an inhibitor of osteoclast attachment in dentinal matrix, *Endod Dent Traumatol* 6(6):255–259, 1990.
121. Wedenberg C, Zetterqvist L: Internal resorption in human teeth—a histological, scanning electron microscopic, and enzyme histochemical study, *J Endod* 13(6):255–259, 1987.
122. Wiener BH, Schilder H: A comparative study of important physical properties of various root canal sealers. II. Evaluation of dimensional changes, *Oral Surg Oral Med Oral Pathol* 32(6):928–937, 1971.
123. Wilson P, Barnes I: Treatment of internal root resorption with thermoplasticized gutta-percha. A case report, *Int Endod J* 20(2):94–97, 1987.
124. Yamaoka M, Furusawa K, Ikeda M, et al: Root resorption of mandibular second molar teeth associated with the presence of the third molars, *Aust Dent J* 44(2):112–116, 1999.
125. Yildirim S, Yapar M, Sermet U, et al: The role of dental pulp cells in resorption of deciduous teeth, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 105(1):113–120, 2008.